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Iwata et al.

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(54) **ROTATING MACHINE**

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H02K 5/173 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **H02K 9/06** (2013.01); **H02K 5/1732** (2013.01); **H02K 5/207** (2021.01); **H02K 7/083** (2013.01); **H02K 2205/09** (2013.01)

(58) **Field of Classification Search**

CPC H02K 9/06; H02K 5/207; H02K 5/1732; H02K 7/083; H02K 2205/09
See application file for complete search history.

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Primary Examiner — Terrance L Kenerly

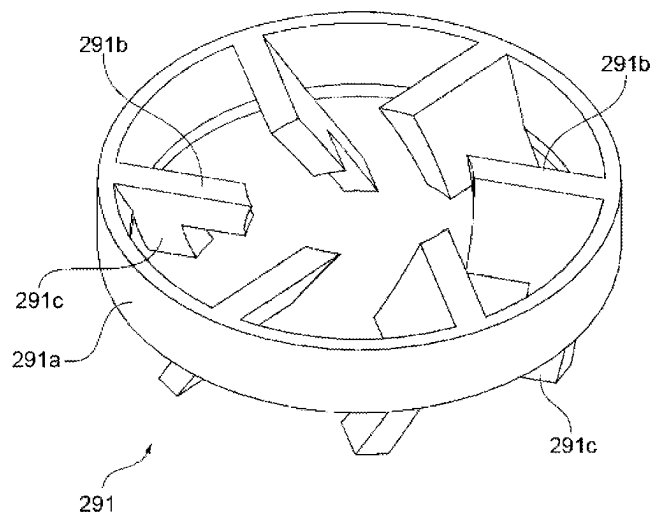
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(57)

ABSTRACT

Provided is a rotating device having excellent performance of cooling a heat generation source. Included are a shaft member, a rotating body having a tubular shape and rotatable about the shaft member, a bearing supporting the rotating body with respect to the shaft member, and a coupling member disposed between the bearing and the rotating body in a radial direction (cd). The coupling member includes a ventilation channel communicating with an inside and an outside of the rotating body.

15 Claims, 25 Drawing Sheets



- (51) **Int. Cl.**
H02K 5/20 (2006.01)
H02K 7/08 (2006.01)

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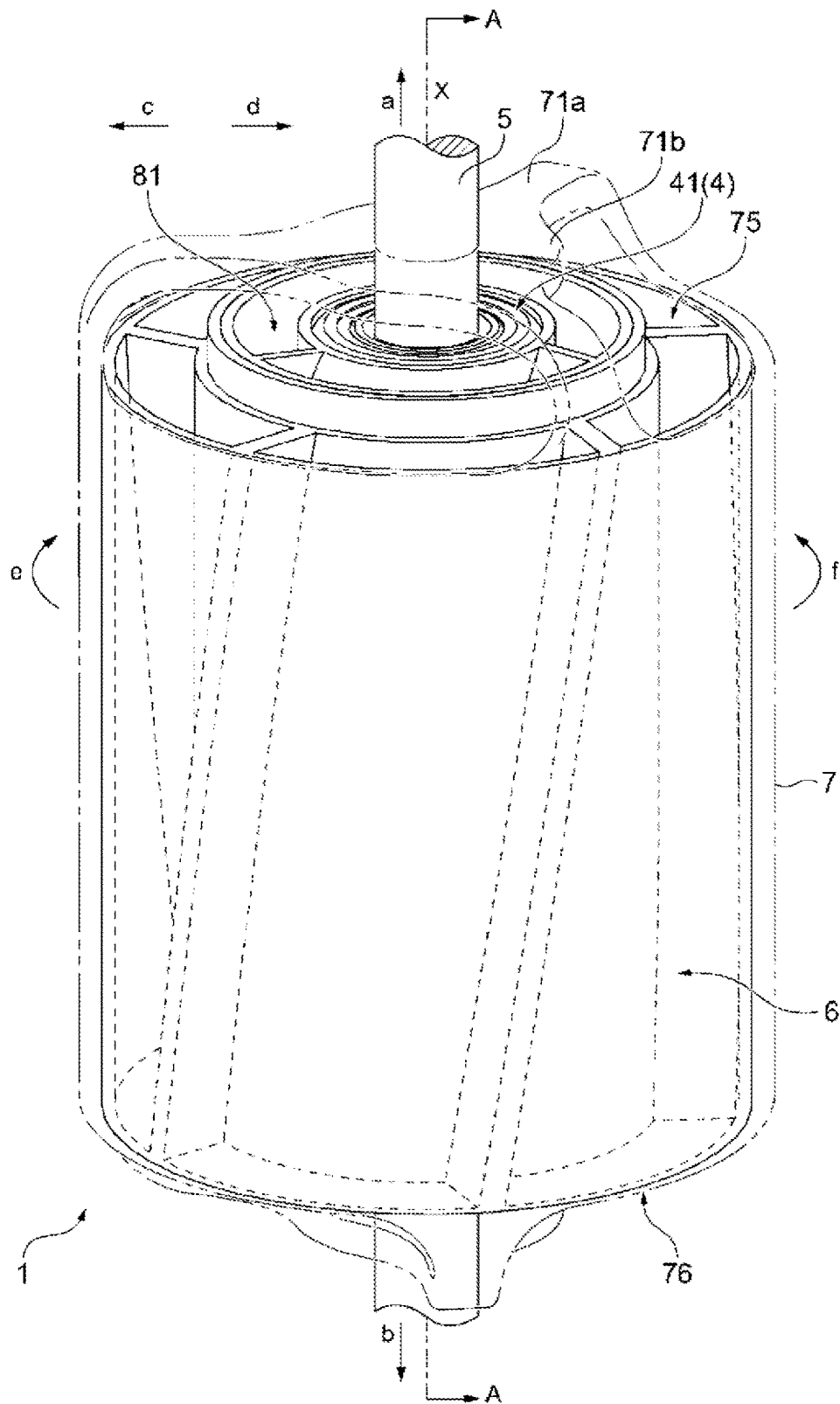


FIG. 1

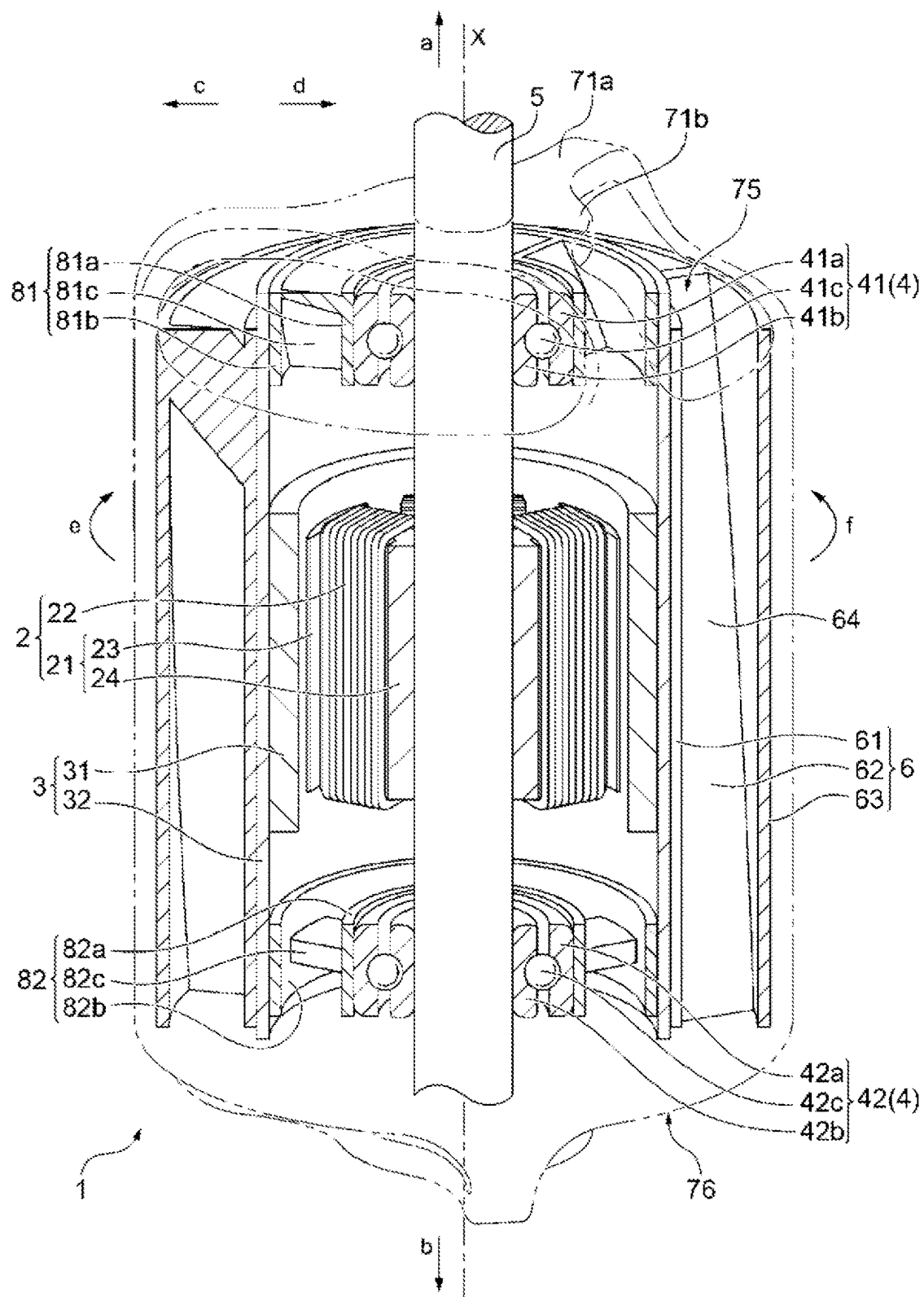


FIG. 2

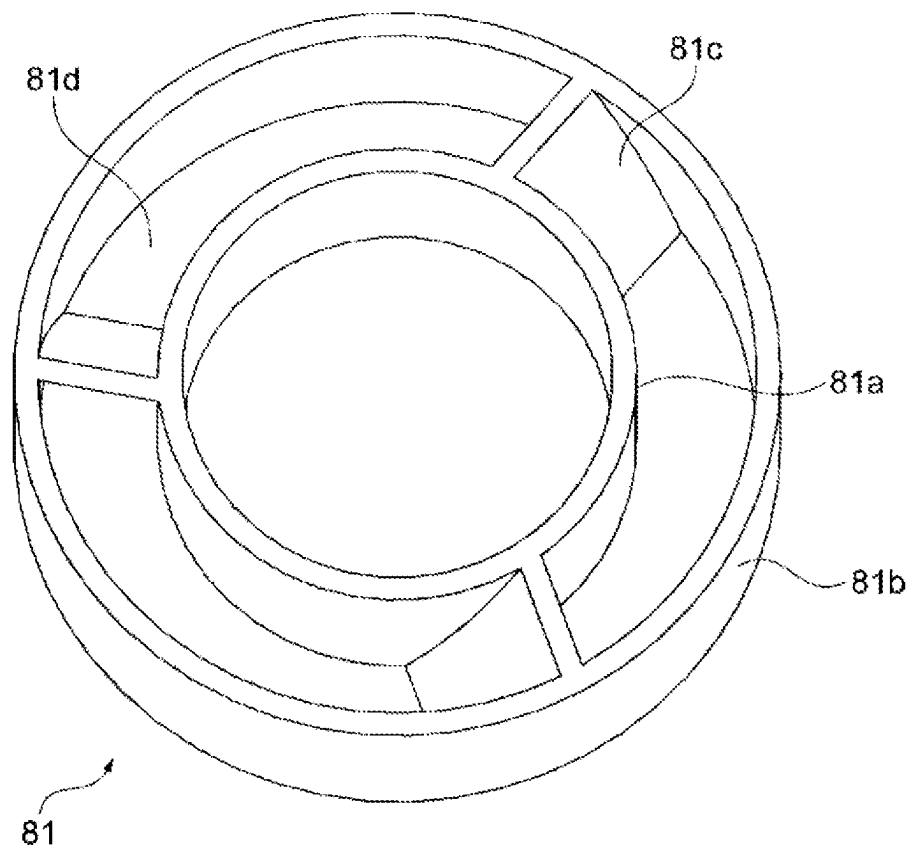


FIG. 3

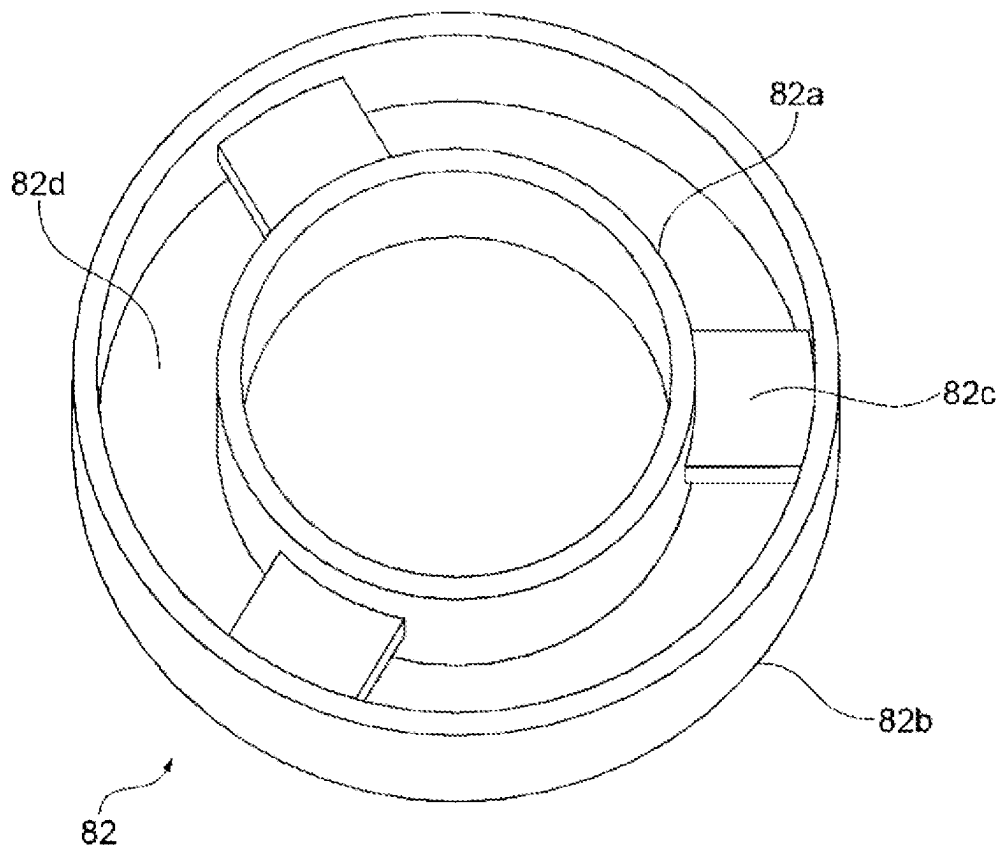


FIG. 4

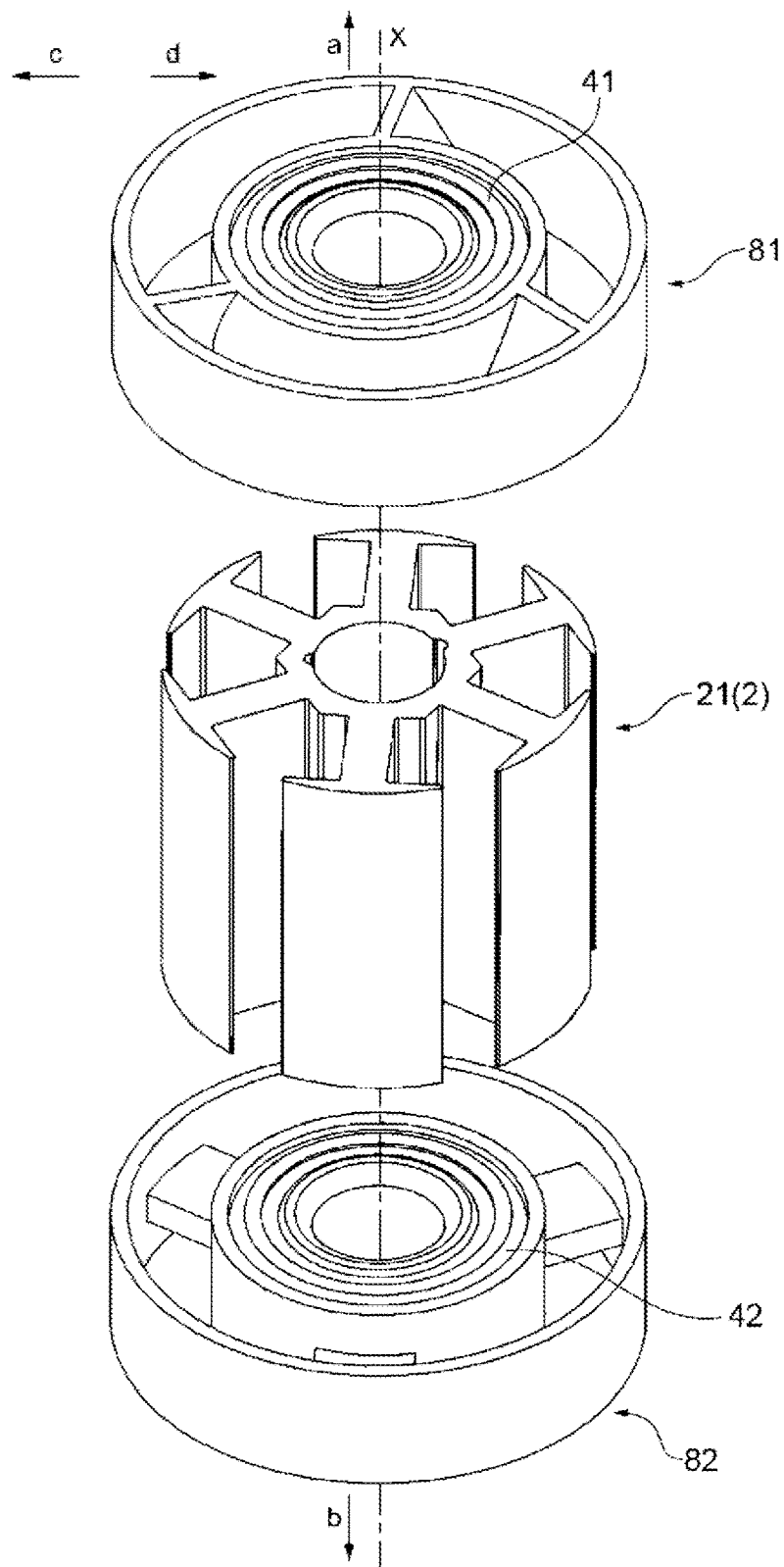


FIG. 5

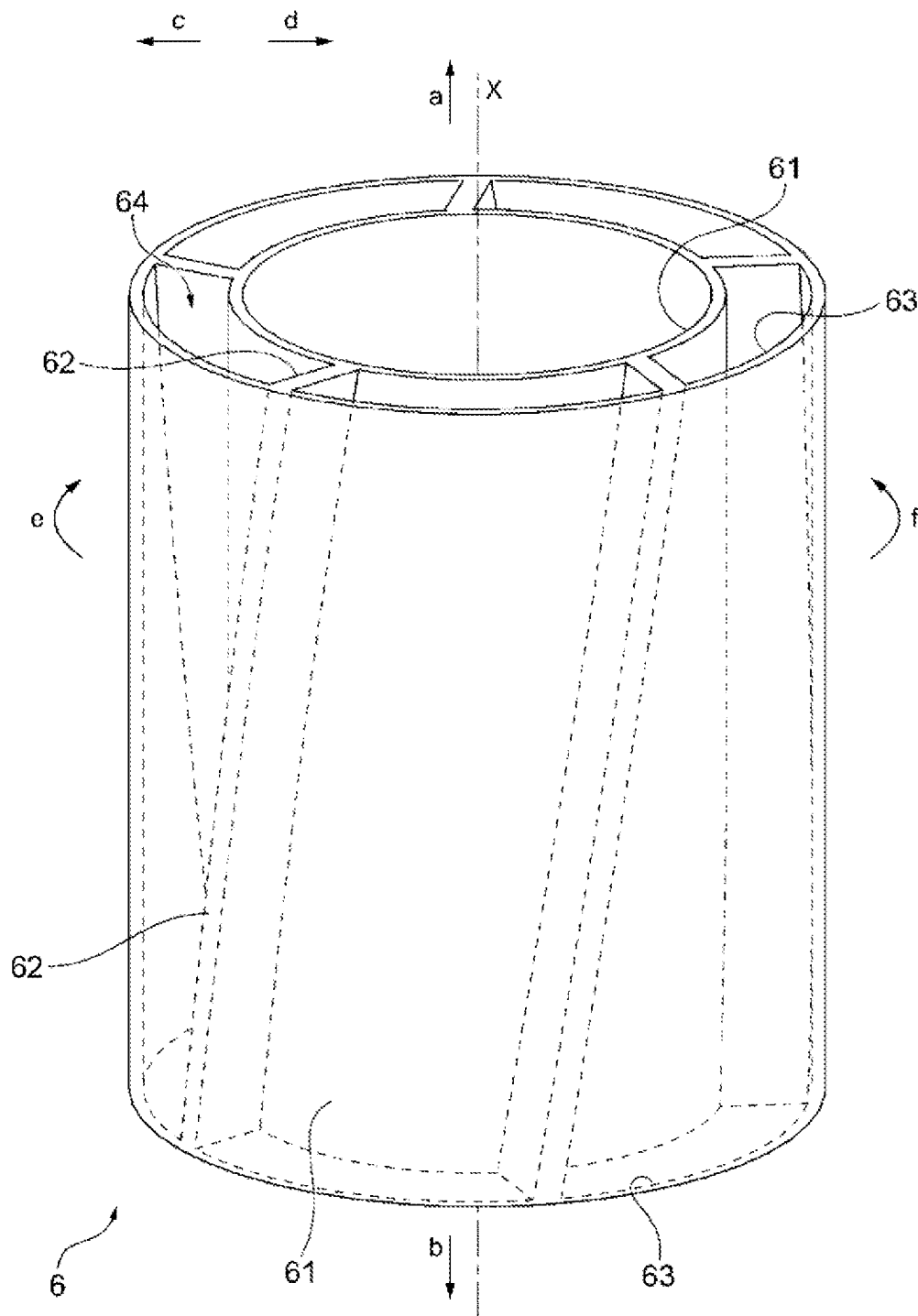


FIG. 6

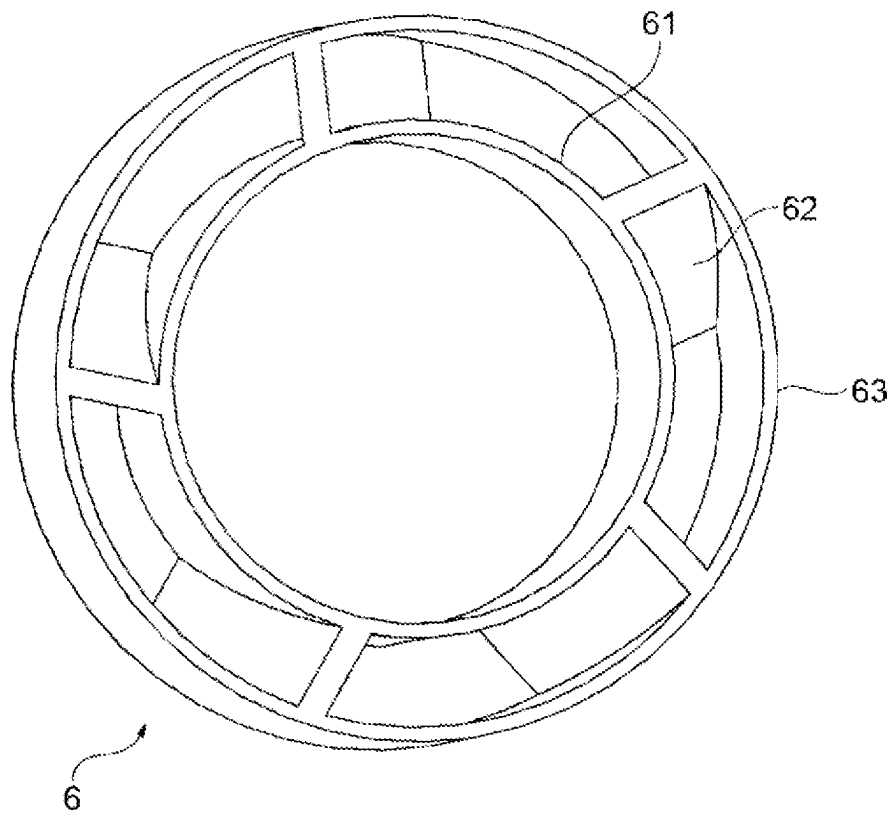


FIG. 7

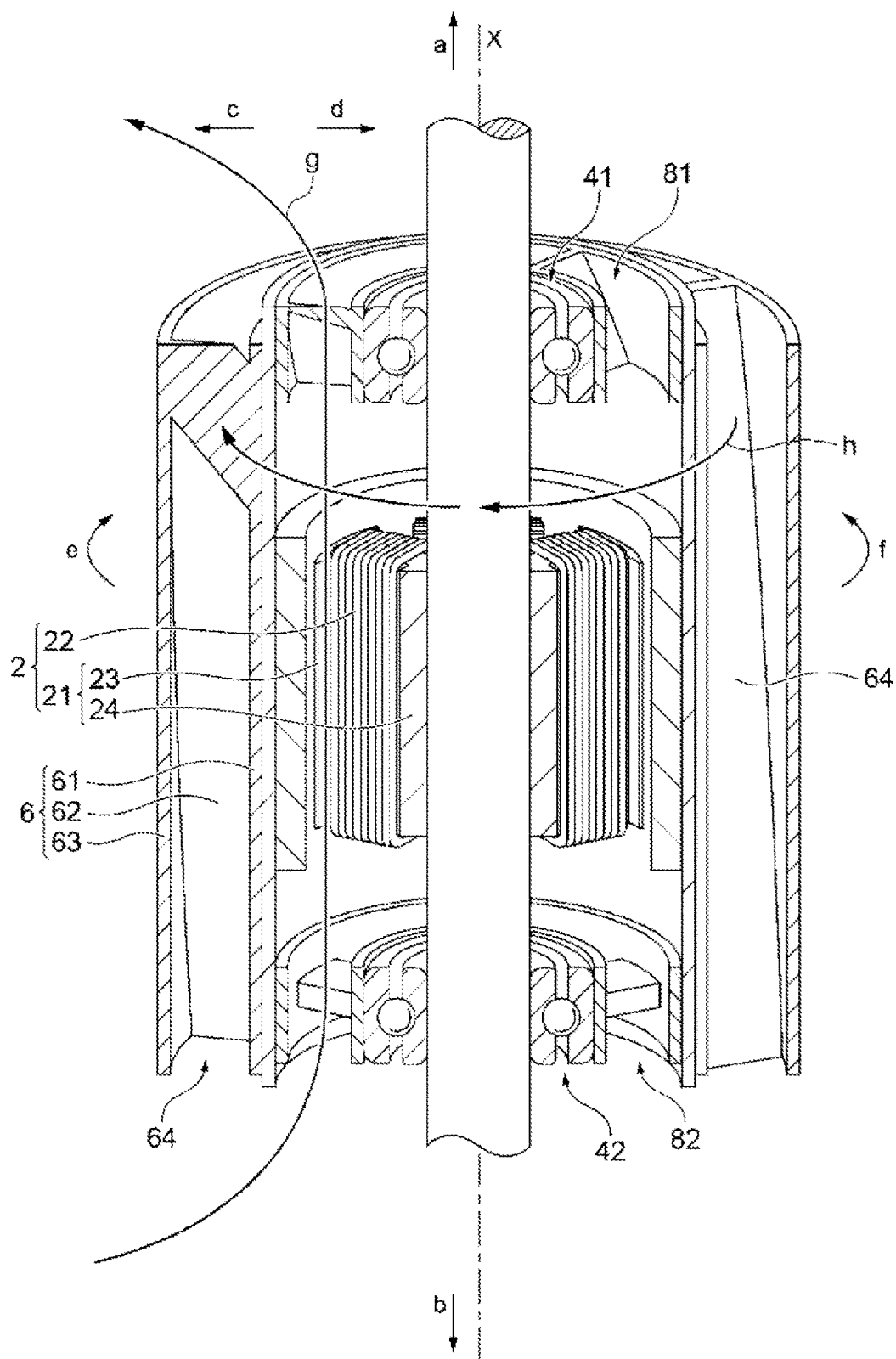


FIG. 8

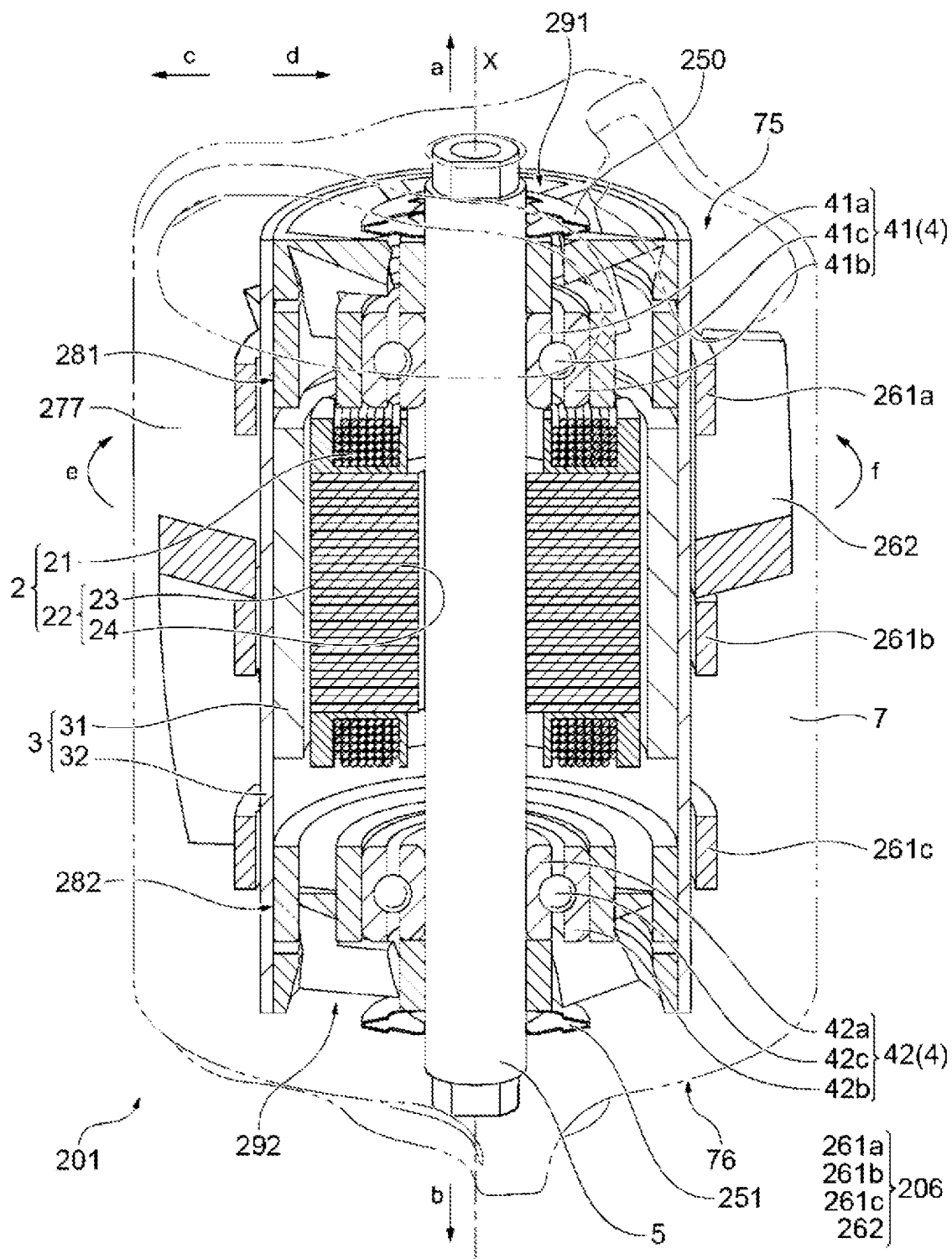


FIG. 9

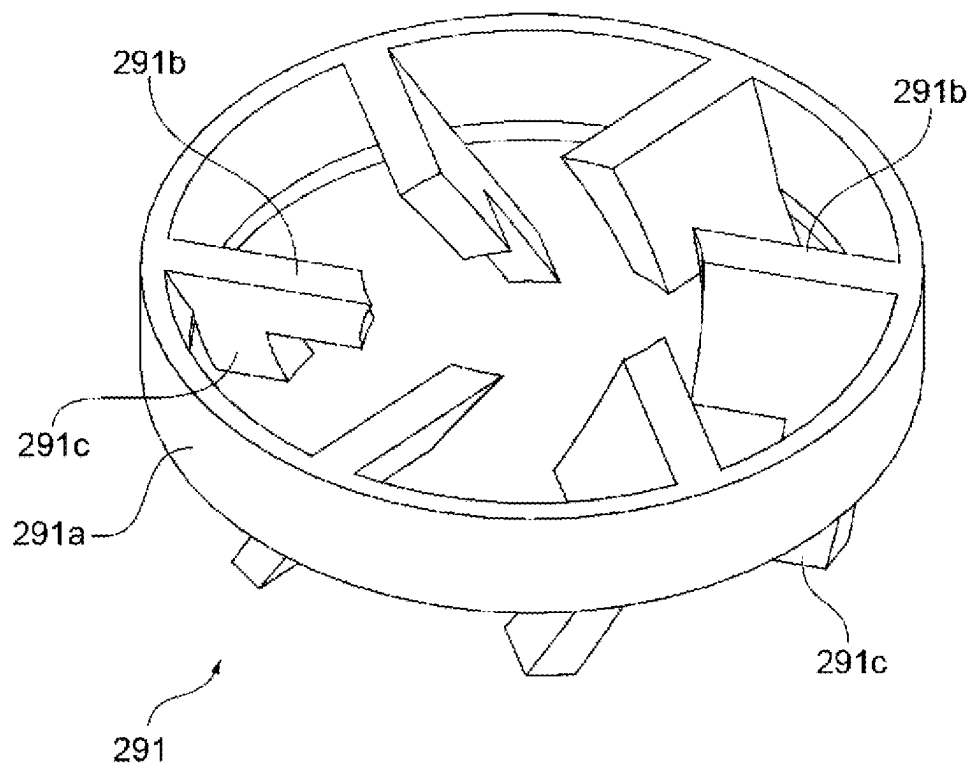


FIG. 10

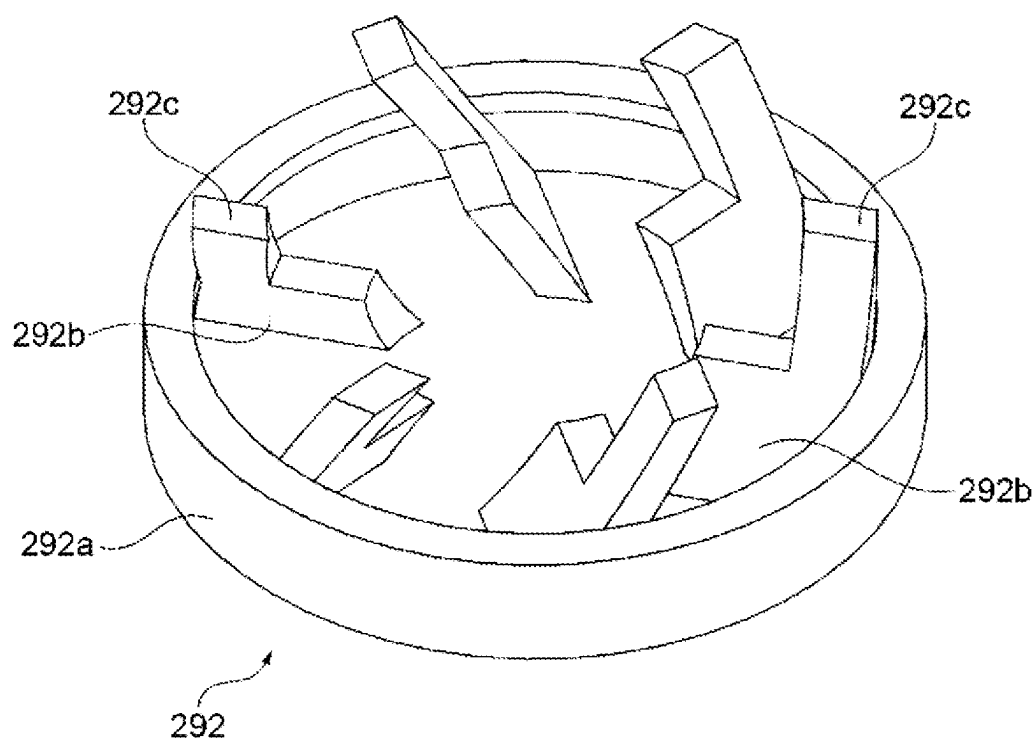


FIG. 11

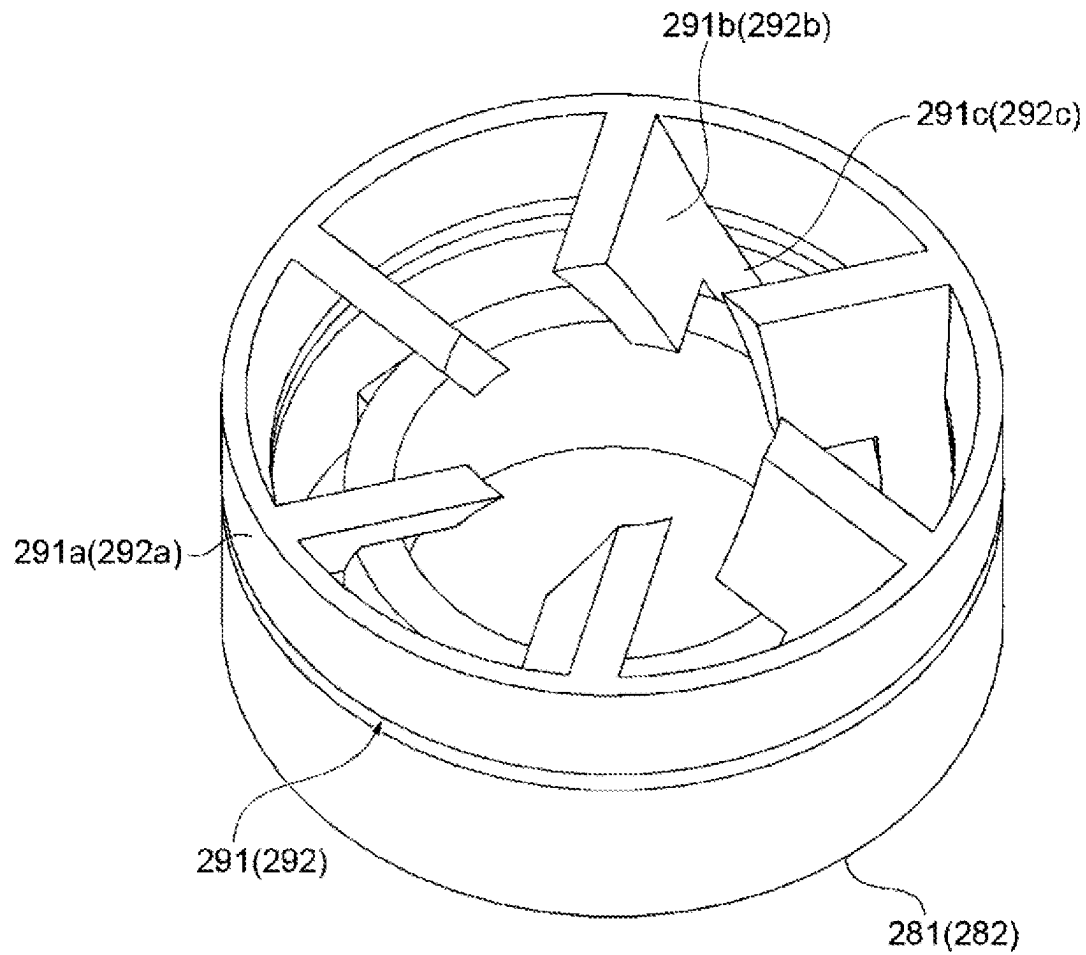


FIG. 12

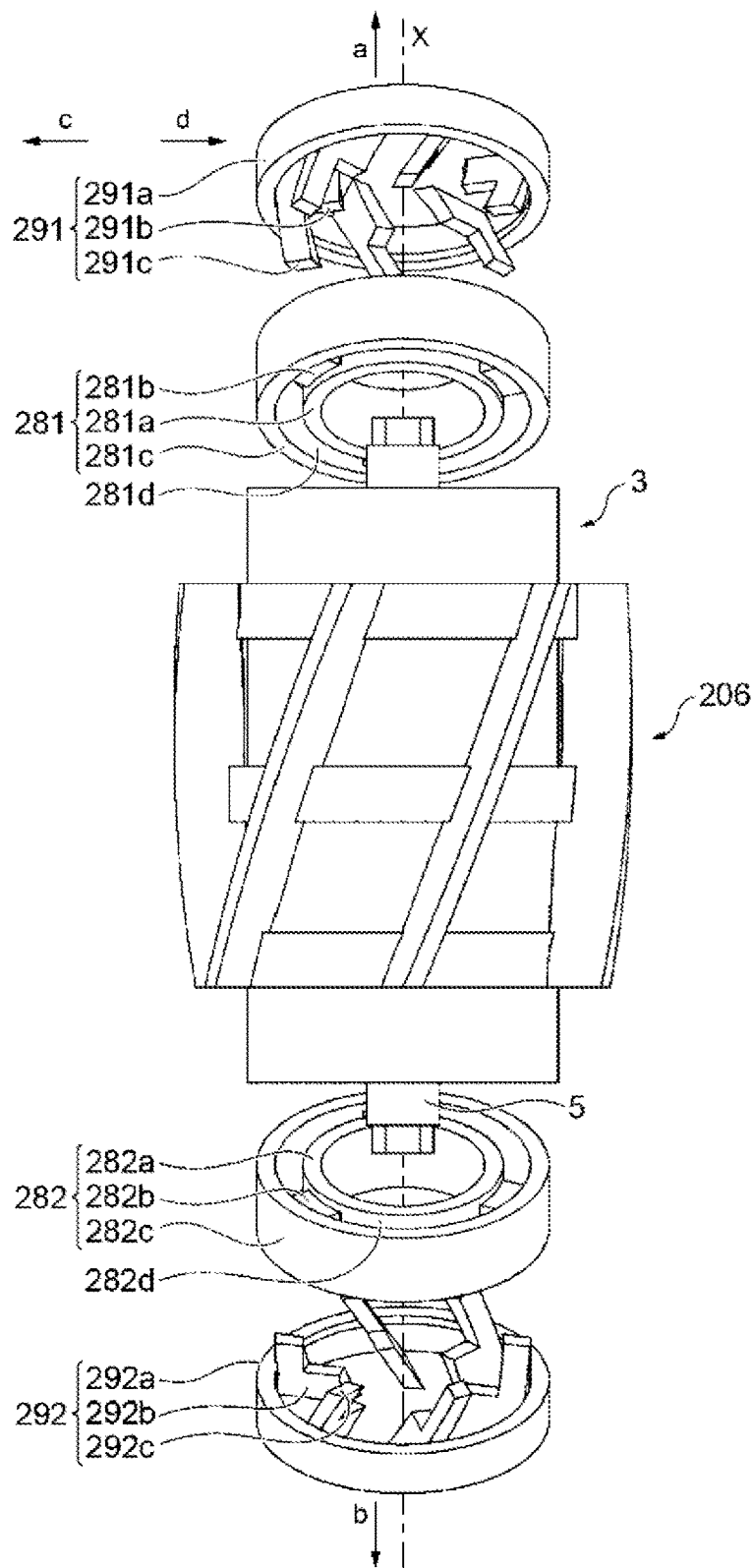


FIG. 13

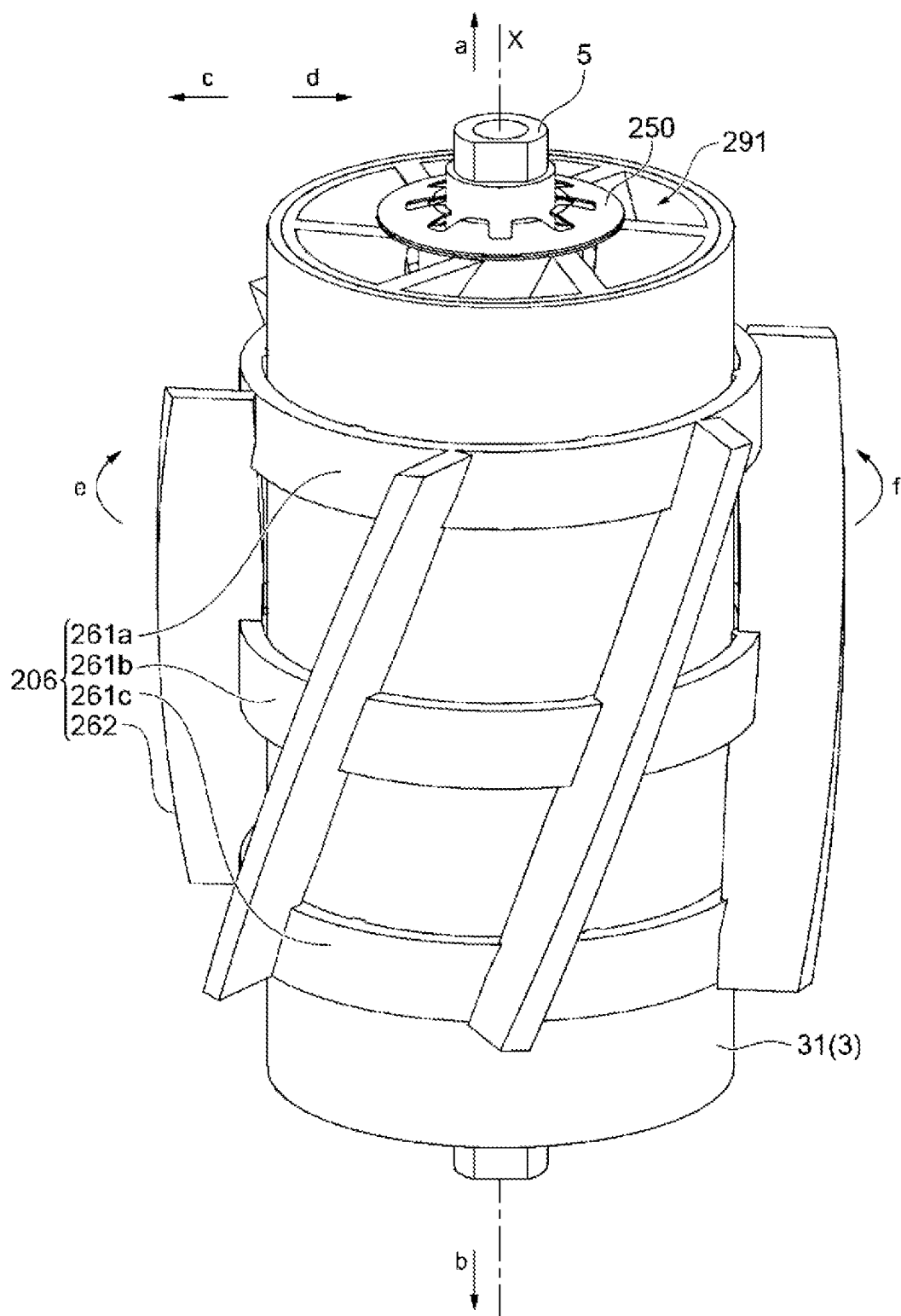


FIG. 14

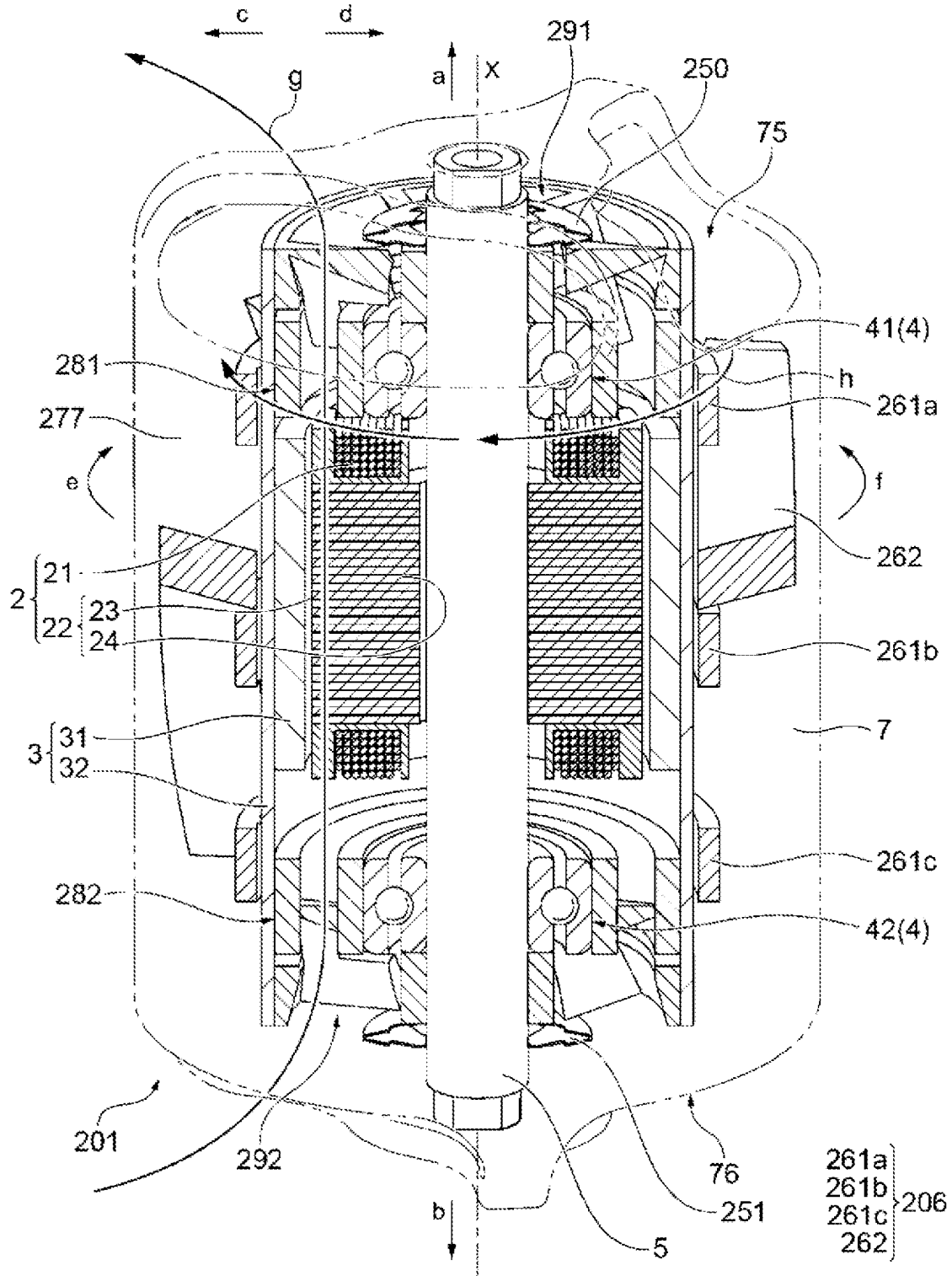


FIG. 15

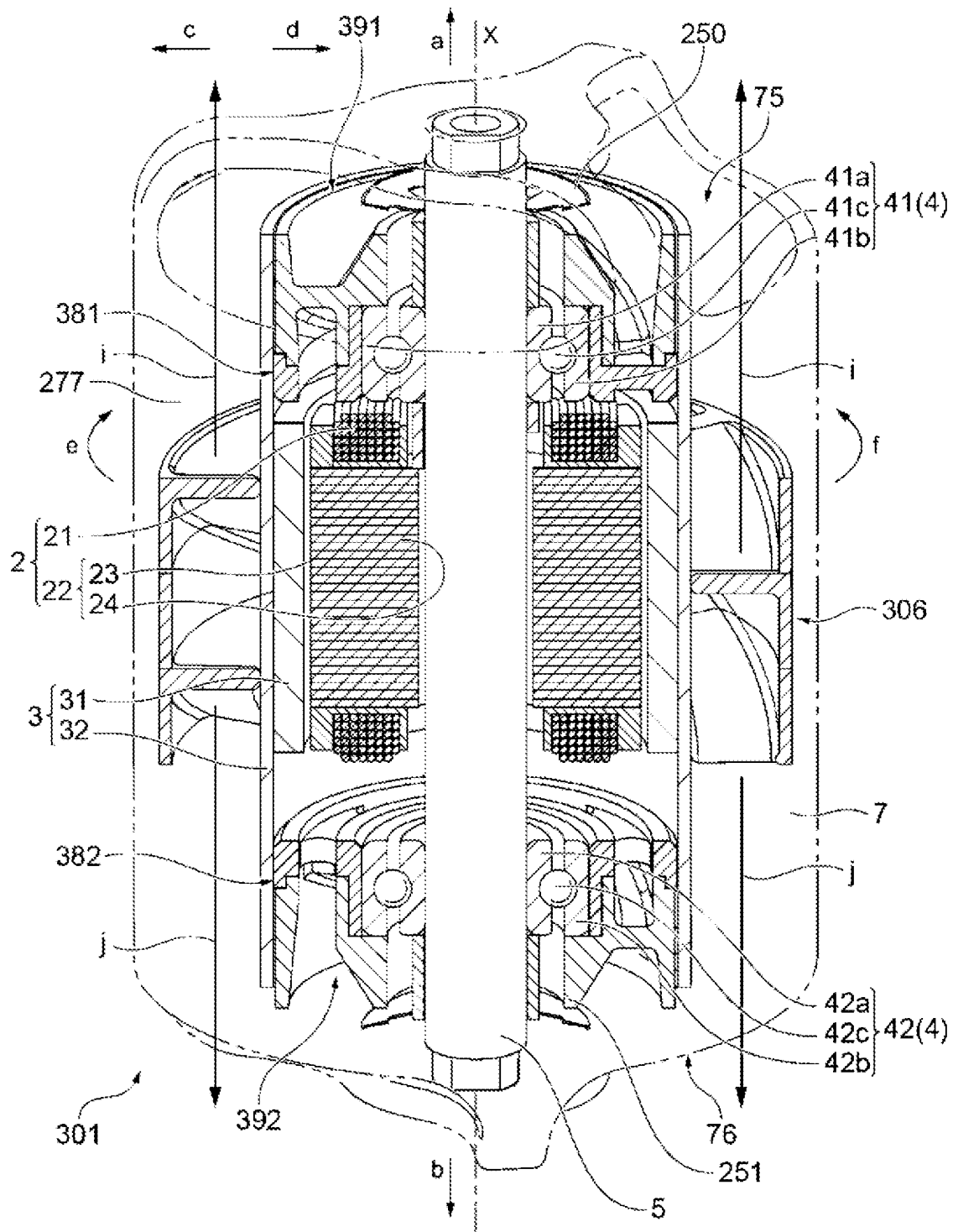


FIG. 16

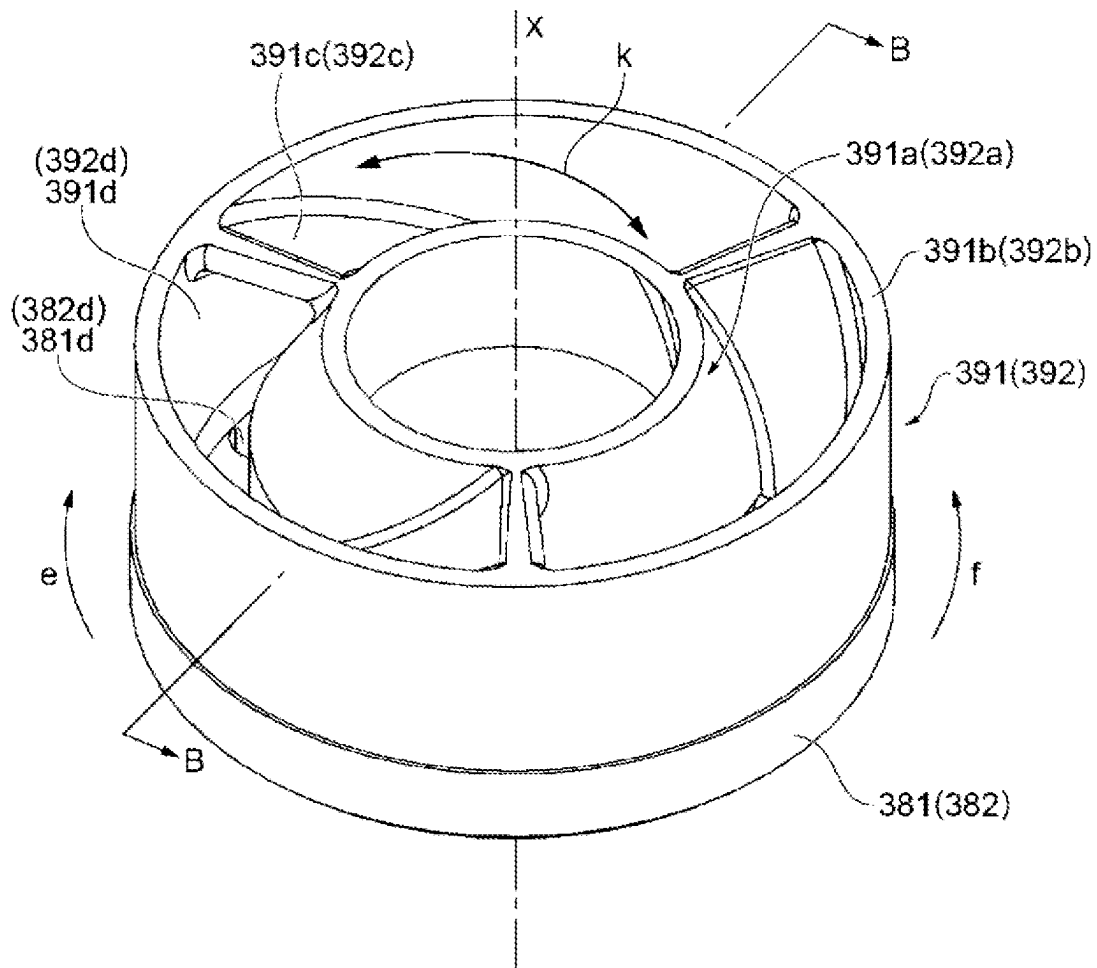


FIG. 17

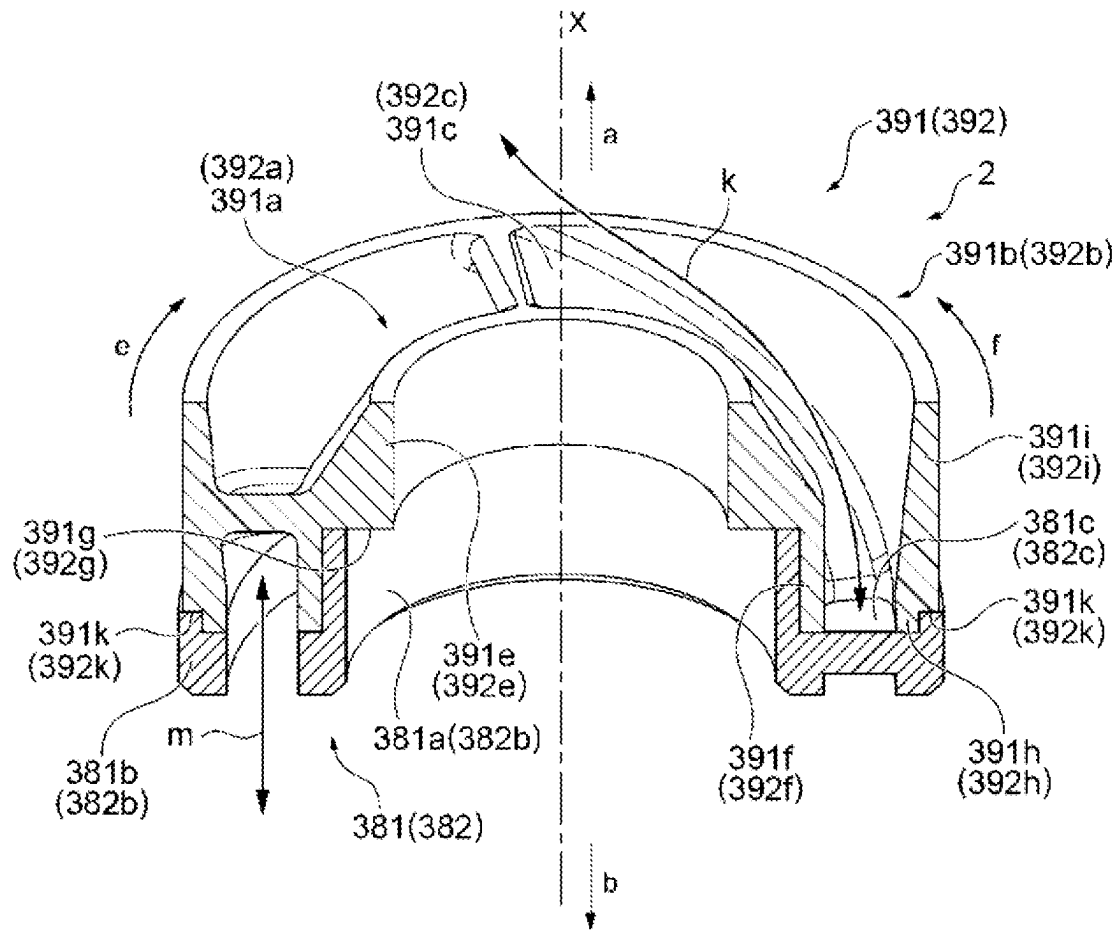


FIG. 18

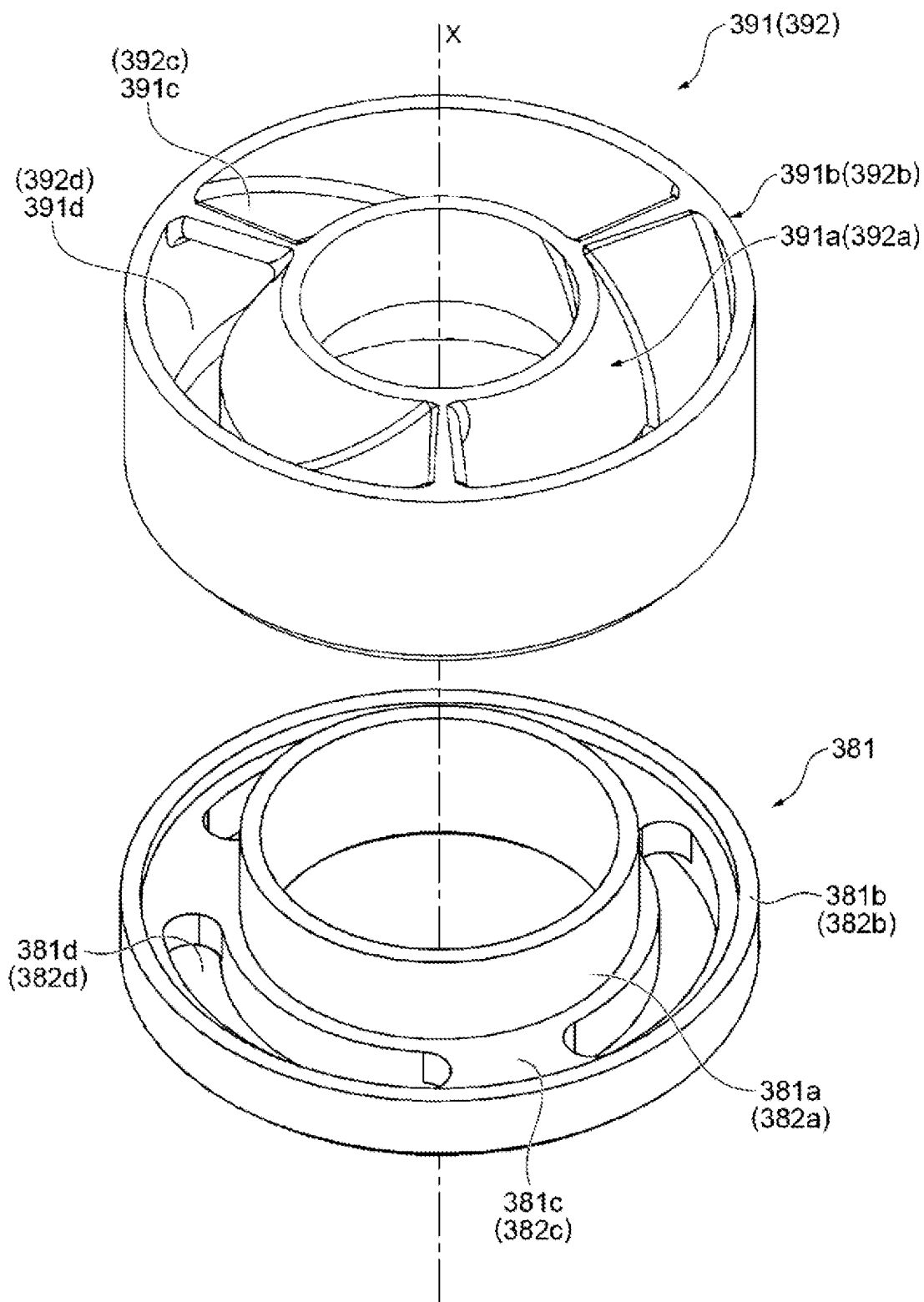


FIG. 19

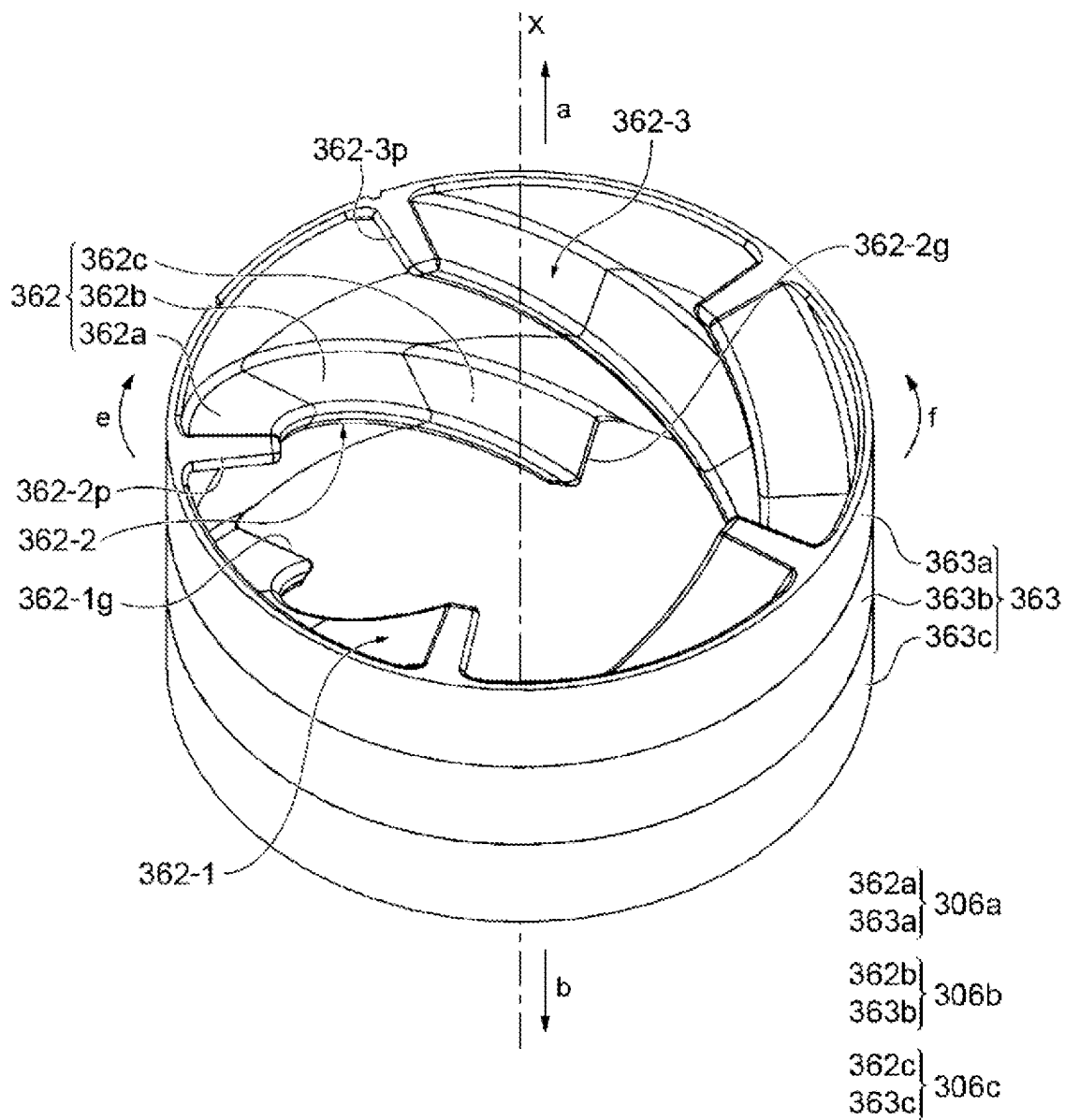


FIG. 20

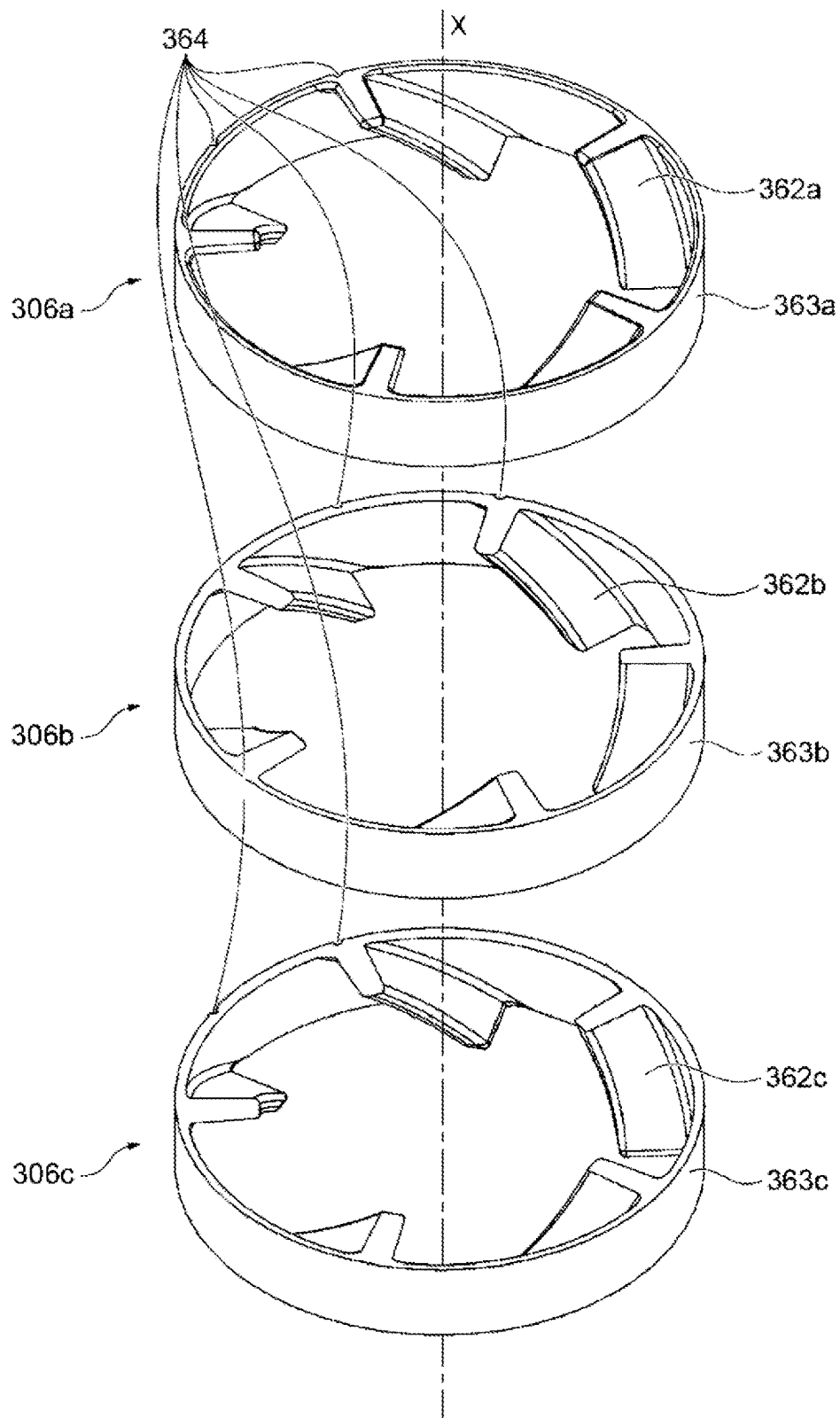


FIG. 21

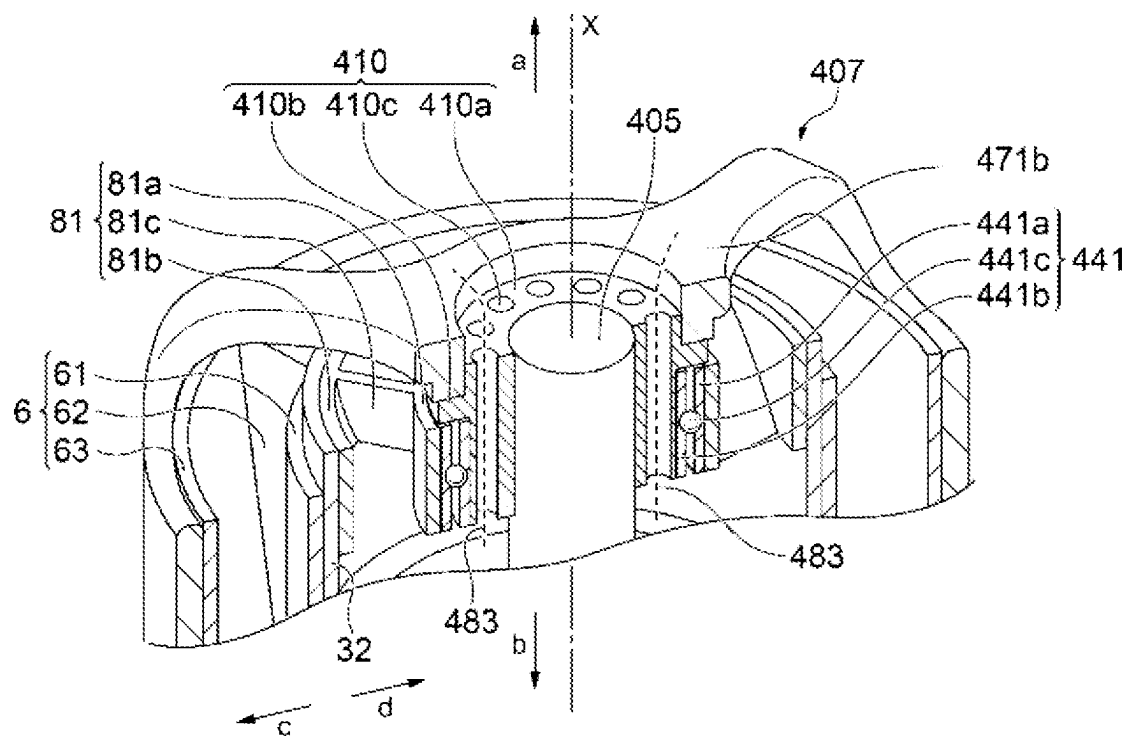


FIG. 22

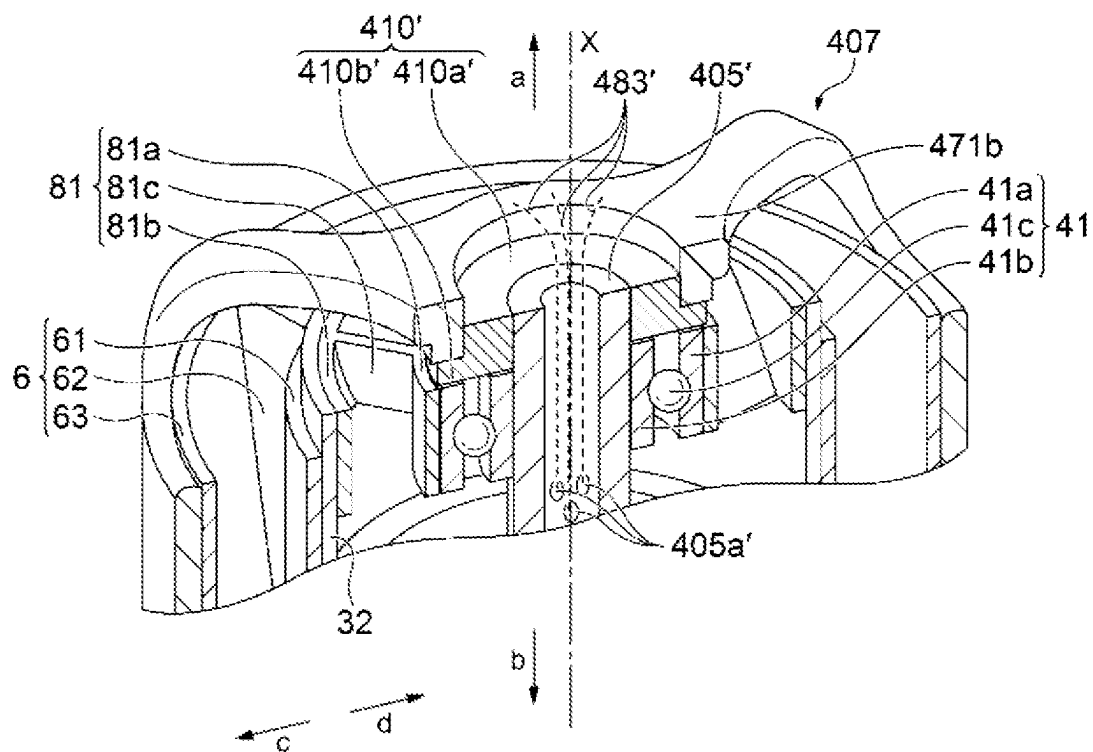


FIG. 23

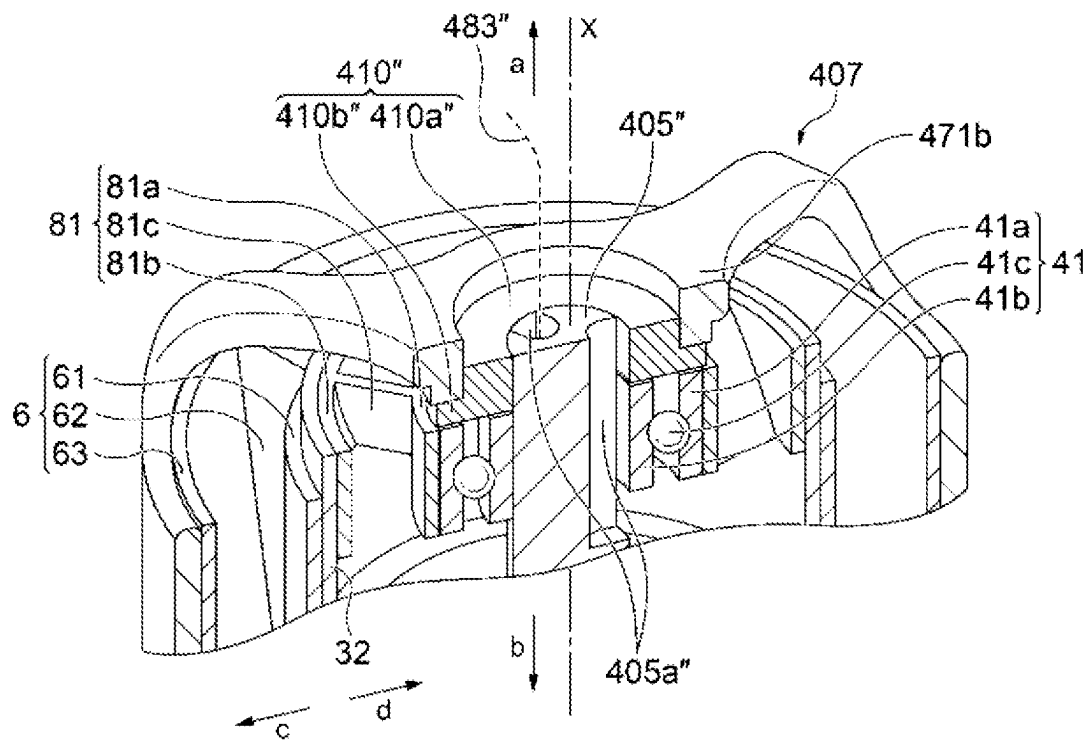


FIG. 24

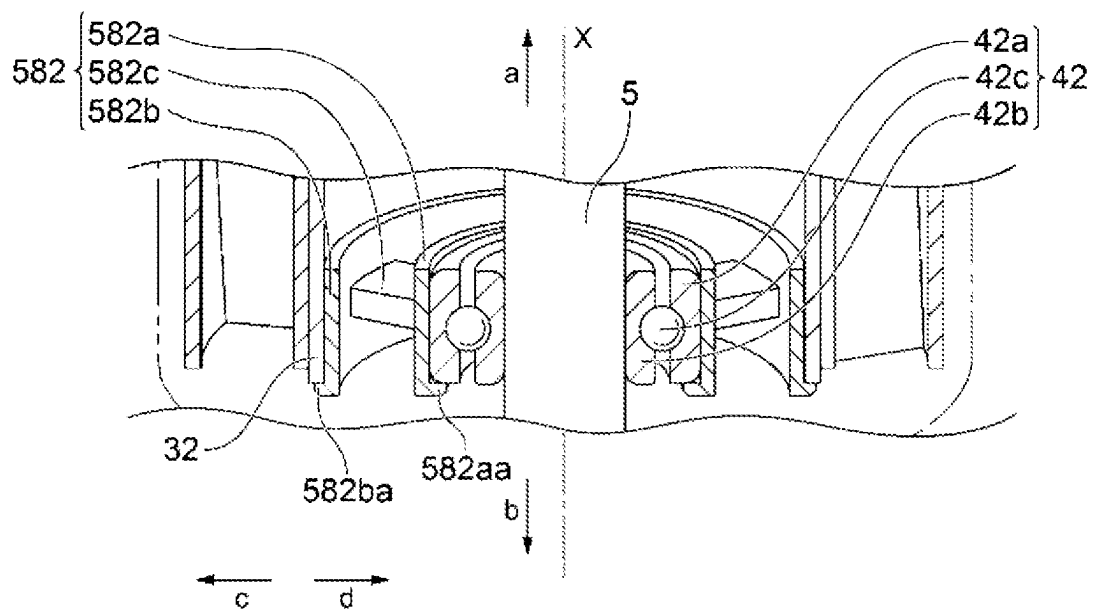


FIG. 25

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ROTATING MACHINE**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a national stage entry of International Application No. PCT/JP2021/030431 filed on Aug. 19, 2021, which claims the benefit of priority to Japanese Application No. JP2020-145023, filed Aug. 28, 2020, the entire disclosures of which are hereby incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a rotating device.

BACKGROUND ART

Known various types of rotating devices have been developed, manufactured, and used in accordance with various applications and required performance. In this context, there is a demand for improving a fundamental performance for a rotating device, such as high speed rotation, and a demand for further size reduction of the entire device, and achieving both the demands at a higher level is awaited.

On the other hand, the high speed rotation causes a coil and a stator core to be more likely to have higher temperature. Thus, there is a need to cool the stator including the heat generation source, such as the coil and the stator core.

CITATION LIST**Patent Literature**

Patent Literature 1: JP 2005-102370 A

SUMMARY OF INVENTION**Technical Problem**

Thus, an example of an object of the present invention is to provide a rotating device having excellent performance of cooling a heat generation source.

Solution to Problem

The above problem is solved by the present invention described below. That is, a rotating device of the present invention includes a shaft member, a rotating body having a tubular shape and rotatable about the shaft member, a bearing supporting the rotating body with respect to the shaft member, and a coupling member disposed between the bearing and the rotating body in a radial direction. The coupling member includes a ventilation channel communicating with an inside and an outside of the rotating body.

The rotating device of the present invention may include a housing having a tubular shape surrounding the rotating body and a stator inside the rotating body.

In the rotating device of the present invention, the coupling member may include a plurality of blades, and the plurality of blades may be oriented in a direction inclined with respect to an axial direction of the shaft member.

In this case, each bearing may be provided at or near a corresponding end part of the rotating body in the axial direction of the shaft member, the coupling member may be disposed between each bearing and the rotating body, and one of each coupling member may include the blades.

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In this case, preferably, a gas can flow inside the rotating body in the axial direction, and at least the coupling member at a discharge port side of the gas includes the blades.

The other of each coupling member may include a plurality of spokes extending inward and outward in the radial direction.

Alternatively, the rotating device of the present invention may include a blade member adjacent to the coupling member in an axial direction and including a plurality of blades, and the blade of the blade member may be inclined.

In this case, the blade of the blade member may be oriented in a direction inclined with respect to an axial direction of the shaft member. Further, the coupling member may include a plurality of spokes extending inward and outward in the radial direction.

In this case, the blade member may include a projecting part projecting toward a side of the coupling member, and the projecting part may fit into the coupling member.

In this case, each bearing may be provided at or near a corresponding end part of the rotating body in an axial direction of the rotating body, the coupling member may be disposed between each bearing and the rotating body, and the blade member may be provided at an outer side of one of each coupling member in the axial direction of the rotating body.

In this case, preferably a gas can flow inside the rotating body in the axial direction, and the blade member is provided at an outer side of the coupling member at a discharge port side of the gas in the axial direction of the rotating body.

Further, the rotating device of the present invention may include a blade member adjacent to the coupling member in an axial direction and including a plurality of blades. The blade member may include a ventilation channel connected to the ventilation channel of the coupling member, and a gas may flow inside the rotating body in both directions of axial direction.

In this case, the coupling member may include a base opposing the blade of the blade member in an axial direction of the shaft member, and a hole part serving as the ventilation channel may be formed in the base.

In the rotating device of the present invention, a rotor blade may be provided at an outer peripheral surface of the rotating body.

In this case, the rotor blade may include a plurality of blades each including a plurality of blade parts lined in the axial direction of the rotating body, and each of the plurality of blades of the rotor blade may oppose a blade adjacent to each of the plurality of blades of the rotor blade in a circumferential direction of the rotating body. An end part of each of the plurality of blades of the rotor blade may be positioned between both end parts of the blade adjacent to each of the plurality of blades of the rotor blade in the circumferential direction of the rotating body.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a transparent perspective view of a rotating device according to a first embodiment as one example of the present invention.

FIG. 2 is a transparent cross-sectional view taken along a cross section A-A in FIG. 1, including an axial line x of the rotating device according to the first embodiment as one example of the present invention.

FIG. 3 is an enlarged perspective view of only a coupling member (inner impeller) including blades extracted from the rotating device according to the first embodiment as one example of the present invention.

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FIG. 4 is an enlarged perspective view of only a coupling member (spacer) including spokes extracted from the rotating device according to the first embodiment as one example of the present invention.

FIG. 5 is an exploded perspective view of only a stator core, a bearing, the inner impeller, and the spacer extracted from the rotating device according to the first embodiment as one example of the present invention.

FIG. 6 is a perspective view of only a rotor blade extracted from the rotating device according to the first embodiment as one example of the present invention.

FIG. 7 is an enlarged perspective view, as viewed from near an upper side a in an axial direction, extracting only the rotor blade from the rotating device according to the first embodiment as one example of the present invention.

FIG. 8 is an explanatory diagram for describing a flow of a gas for cooling an inside of a rotor of the rotating device according to the first embodiment as one example of the present invention.

FIG. 9 is a transparent cross-sectional view of a cross section including an axial line x of a rotating device according to a second embodiment as one example of the present invention.

FIG. 10 is an enlarged perspective view of only a blade member at an upper side a in FIG. 9 extracted from the rotating device according to the second embodiment as one example of the present invention.

FIG. 11 is an enlarged perspective view of only a blade member at a lower side b in FIG. 9 extracted from the rotating device according to the second embodiment as one example of the present invention.

FIG. 12 is an enlarged perspective view of only the blade member and the spacer at the upper side a (lower side b) in FIG. 9 extracted from the rotating device according to the second embodiment as one example of the present invention.

FIG. 13 is an exploded perspective view of the rotating device according to the second embodiment as one example of the present invention.

FIG. 14 is a perspective view with only a housing removed from the rotating device according to the second embodiment as one example of the present invention.

FIG. 15 is an explanatory diagram for describing a flow of a gas for cooling an inside of a rotor of the rotating device according to the second embodiment as one example of the present invention.

FIG. 16 is a transparent cross-sectional view of a cross section including an axial line x of a rotating device according to a third embodiment as one example of the present invention.

FIG. 17 is an enlarged perspective view of only a spacer and a blade member at an upper side a (lower side b) in FIG. 16 extracted from the rotating device according to the third embodiment as one example of the present invention.

FIG. 18 is a cross-sectional perspective view taken along a cross section B-B including the axial line x in FIG. 17.

FIG. 19 is an exploded perspective view of only the spacer and the blade member extracted from the rotating device according to the third embodiment as one example of the present invention.

FIG. 20 is a perspective view of only rotor blades extracted from the rotating device according to the third embodiment as one example of the present invention.

FIG. 21 is an exploded perspective view of only the rotor blades extracted from the rotating device according to the third embodiment as one example of the present invention.

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FIG. 22 is an enlarged perspective cross-sectional view of a rotating device according to a modification example of the present invention.

FIG. 23 is an enlarged perspective cross-sectional view of a rotating device according to a modification example of the present invention.

FIG. 24 is an enlarged perspective cross-sectional view of a rotating device according to a modification example of the present invention.

FIG. 25 is an enlarged cross-sectional view of a rotating device according to a modification example of the present invention.

DESCRIPTION OF EMBODIMENTS

Rotating devices according to embodiments of the present invention will be described below with reference to the drawings. In the following embodiments, the rotating devices will be described with an example of a so-called blower (or intake device) configured to generate wind for the purpose of blowing (or intaking) air.

First Embodiment

FIG. 1 is a cross-sectional view of a rotating device 1 according to a first embodiment as one example of the present invention, and FIG. 2 is a transparent cross-sectional view taken along a cross section (cross-section A-A in FIG. 1), including an axial line x of the rotating device 1. FIGS. 1 and 2 illustrates a housing 7 in a transparent state by being drawn with imaginary lines (chain double-dashed lines).

In the description of the first embodiment and a second embodiment described below, “upper” or “lower” refers to a vertically structured relationship in FIGS. 1 and 2, and do not necessarily match a vertically structured relationship in the direction of gravity.

In the axial line x direction (hereinafter also referred to as the “axial direction”), an arrow a direction is referred to as an upper side a, and an arrow b direction is referred to as a lower side b. Further, in a direction perpendicular to the axial line x (hereinafter also referred to as a “radial direction”), a direction away from the axial line x (arrow c direction) is referred to as an outer peripheral side c, a direction toward the axial line x (arrow d direction) is referred to as an inner peripheral side d, and both directions are referred to as radial directions cd. A clockwise direction in a circumferential direction (circumferential direction viewed from the upper side a) about the axial line x is referred to as a circumferential direction e, and a counterclockwise direction in the circumferential direction is referred to as a circumferential direction f. The arrows a to f indicating these directions and x indicating the axial line are similarly applied to the description of the second embodiment described below.

In addition, in the description of the present embodiment, in the rotating device 1, a part rotating may be referred to as a “rotating side”, and a part supporting a member at the rotating side and fixed without rotating may be referred to as a “fixed side”. Since the part fixed without rotating is relatively stationary with respect to the part rotating, the part fixed without rotating may be referred to as a stationary part. These terms are similarly applied to the description of the second embodiment described below.

The rotating device 1 of the present embodiment is an outer-rotor type rotating device including a shaft member 5, a rotor 3 as a rotating body having a tubular shape and rotatable about the shaft member 5, the housing 7 having a tubular shape and surrounding the rotor 3, a bearing 4

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rotatably supporting the rotor 3 about the shaft member 5, a coupling member 81, 82 disposed between the rotor 3 and the bearing 4 and coupling the rotor 3 with the bearing 4, a stator 2 inside the rotor 3, and a plurality of rotor blades 6 provided on the rotor 3.

The stator 2 includes a stator core 21 and a coil 22. The stator core 21 is fixed to the shaft member 5 and includes a plurality of magnetic pole parts 23 extending radially toward the outer peripheral side from the shaft member 5 as an axis. The coil 22 is wound around the magnetic pole parts 23. The illustrated stator 2 is disposed in the housing 7 so that a gap between a first bearing 41 and the stator 2 is equal to a gap between a second bearing 42 and the stator 2.

The stator core 21 includes an annular part 24 and the plurality of magnetic pole parts 23. The annular part 24 is a stacked body formed by stacking magnetic bodies such as silicon steel plates and is disposed coaxially surrounding the shaft member 5. The plurality of magnetic pole parts 23 is formed extending radially toward the outer peripheral side in the radial direction from the annular part 24.

The rotor 3 includes a magnet 31 and a tubular member 32. The magnet 31 opposes the magnetic pole parts 23 at the outer peripheral side of the stator 2. The tubular member 32 has a tubular shape. The magnet 31 is disposed at an inner peripheral surface of the tubular member 32. The tubular member 32 has a cylindrical shape centered at an axis of the shaft member 5 and surrounds the stator 2. The tubular member 32 also has a function of preventing leakage of a magnetic field from an interior of the tubular member 32 and is formed of a magnetic material. Note that the tubular member 32 may be formed of a non-magnetic material such as aluminum or plastic, for example, as long as there is no problem with the characteristics of the tubular member 32.

The magnet 31 is attached to the inner peripheral surface of the tubular member 32 so as to oppose the stator 2. The magnet 31 has an annular shape, and is provided with a region magnetized to the north pole and a region magnetized to the south pole alternately at a regular cycle (or at regular intervals) along a circumferential direction. The magnet 31 may be an annular integrally molded article; however, a plurality of magnets may be attached in a row to the inner peripheral surface of the tubular member 32 and arranged in a tubular shape.

The bearings 4 are disposed at both sides of the stator 2 in the axial direction of the shaft member 5 and includes two bearings, the first bearing 41 positioned at the upper side a and the second bearing 42 positioned at the lower side b. In other words, the magnet 31 and the stator 2 are positioned between the first bearing 41 and the second bearing 42 in the axial direction of the shaft member 5. The first bearing 41 and the second bearing 42 are members having the same configuration (same shape, structure, size, and material).

The first bearing 41 is a so-called ball bearing including an outer ring 41a, an inner ring 41b, and bearing balls 41c interposed between the outer ring 41a and the inner ring 41b. The bearing balls 41c roll between the outer ring 41a and the inner ring 41b, significantly reducing a rotational resistance of the outer ring 41a with respect to the inner ring 41b. The first bearing 41 is formed, for example, of a hard metal such as iron, or a ceramic in consideration of its function. The same applies to the second bearing 42.

The coupling member (hereinafter referred to as "inner impeller") 81 is disposed between the outer ring 41a of the first bearing 41 and the inner peripheral surface of an end part of the tubular member 32 at the upper side a. FIG. 3 illustrates an enlarged perspective view of only the inner impeller 81 extracted. As illustrated in FIG. 3, the inner

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impeller 81 includes a ring part (hereinafter referred to as "inner ring part") 81a at an inner side, a ring part (hereinafter referred to as "outer ring part") 81b at an outer side, and a plurality of (three in the present embodiment) blades 81c connecting an area between the inner ring part 81a and the outer ring part 81b.

In the inner impeller 81, a gas can pass between the inner ring part 81a and the outer ring part 81b, excluding positions of the blades 81c. That is, the inner impeller 81 includes a ventilation channel 81d.

The inner impeller 81 may be molded with any material such as a resin, aluminum, or other metals but is preferably molded with a resin from the perspectives of weight reduction, low cost, and moldability.

As illustrated in FIG. 3, a plurality of blades 81c is oriented in a direction inclined with respect to the axial line x direction (axial direction of the shaft member 5). The three blades 81c are inclined at the same angle. Therefore, the inner impeller 81 functions as a so-called fan for generating a flow of a gas in the axial line x direction by rotating.

On the other hand, the coupling member (hereinafter referred to as "spacer") 82 is disposed between an outer ring 42a of the second bearing 42 and the inner peripheral surface of an end part of the tubular member 32 at the lower side b. FIG. 4 illustrates an enlarged perspective view of only the spacer 82 extracted. As illustrated in FIG. 4, the spacer 82 includes a ring part (hereinafter referred to as "inner ring part") 82a at an inner side, a ring part (hereinafter referred to as "outer ring part") 82b at an outer side, and a plurality of (three in the present embodiment) spokes 82c connecting an area between the inner ring part 82a and the outer ring part 82b.

The spacer 82 forms a ventilation channel 82d enabling the passage of a gas at positions excluding positions of the spokes 82c between the inner ring part 82a and the outer ring part 82b.

The spacer 82 may be molded with any material such as a resin or aluminum or other metal, but is preferably molded with a resin from the perspectives of weight reduction, low cost, and moldability.

As illustrated in FIG. 4, a plurality of spokes 82c in the spacer 82 has a side surface (plate surface) facing toward the stator 2 not inclined with respect to the axial line x direction, unlike the plurality of blades 81c of the inner impeller 81.

An inner peripheral surface of the inner ring part 81a of the inner impeller 81 is fixed to an outer peripheral surface of the outer ring 41a of the first bearing 41. On the other hand, an inner peripheral surface of the inner ring part 82a of the spacer 82 is fixed to an outer peripheral surface of the outer ring 42a of the second bearing 42. The fixing of the inner peripheral surfaces of the inner ring parts 81a, 82a and the outer peripheral surfaces of the outer rings 41a, 42a is not particularly limited, and may be performed by any known method, such as, for example, press-fitting, interference-fitting, fixing with an adhesive, or engagement, or a combination of a plurality of these fixing means (fixing elements).

FIG. 5 illustrates an exploded perspective view of only the stator core 21, the bearing 4, the inner impeller 81, and the spacer 82 of the rotating device 1 according to the present embodiment extracted.

The inner ring part 81a of the inner impeller 81 is fixed to the outer ring 41a of the first bearing 41. The inner ring part 82a of the spacer 82 is fixed to the outer ring 42a of the second bearing 42.

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In FIG. 5, only the stator core 21 without illustration of the coil 22 is depicted; however, actually the stator 2 with the coil 22 wound around the magnetic pole parts 23 of the stator core 21 is prepared.

As illustrated in FIG. 5, the shaft member 5 is inserted and fitted into the center axis x in the order of an assembly of the first bearing 41 and the inner impeller 81, the stator 2, and the second bearing 42 and the spacer 82.

The inner ring 41b of the first bearing 41 and the inner ring 42b of the second bearing 42 are loosely fitted to the shaft member 5, and then fixed by an adhesive. Thus, a gap between the inner ring 41b and the shaft member 5, and a gap between the inner ring 42b and the shaft member 5 are filled with the adhesive. Alternatively, the inner ring 41b of the first bearing 41 and the inner ring 42b of the second bearing 42 may be press-fitted and fixed to the shaft member 5.

Thus, the inner ring 41b of the first bearing 41 and the inner ring 42b of the second bearing are fixed with respect to the shaft member 5 and serve as stationary parts together with the shaft member 5. Here, the shaft member 5 and the housing 7 are members stationary with respect to (relative to) the rotor 3. Thus, these are collectively referred to as a stationary member (stationary part).

On the other hand, an outer peripheral surface of the outer ring part 81b of the inner impeller 81 is fixed to the inner peripheral surface of the end part of the tubular member 32 at the upper side a. An outer peripheral surface of the outer ring part 82b of the spacer 82 is fixed to the inner peripheral surface of the end part of the tubular member 32 at the lower side b. The fixing of the outer peripheral surfaces of the outer ring parts 81b, 82b and the inner peripheral surface of the tubular member 32 is also not particularly limited, and may be performed by any known method, such as, for example, press-fitting, interference-fitting, fixing with an adhesive, or engagement, or a combination of a plurality of these fixing means (fixing elements).

Accordingly, the inner ring 41b of the first bearing 41 and the inner ring 42b of the second bearing 42 are fixed to an outer peripheral surface of the shaft member 5, and the outer ring 41a of the first bearing 41 and the outer ring 42a of the second bearing 42 are fixed to the inner peripheral surface at both ends of the tubular member 32 with the inner impeller 81 and the spacer 82 interposed between the outer rings 41a, 42a and the inner peripheral surface of the tubular member 32. As described above, the rotor 3 is configured to be rotatable about the axial line x of the shaft member 5 as a center axis.

The shaft member 5 is formed of aluminum, for example, into a hollow state (more specifically, a cylindrical state) for weight reduction. In the present embodiment, the shaft member 5 is a member at the fixed side. Since the member has a function of supporting the stator 2, the rotor 3, the bearing 4, and the rotor blades 6 with respect to the housing 7, it is necessary to have rigidity corresponding to the function.

An opening (not illustrated) is provided at the middle (intermediate part) of the shaft member 5, and a lead wire (not illustrated) connected to the coil 22 is drawn from the opening into a cavity within the shaft member 5, and is pulled out of the rotating device 1 from an end part opening (not illustrated) of the shaft member 5.

In the rotating device 1 according to the present embodiment, the first bearing 41 and the second bearing 42 as well as the inner impeller 81 and the spacer 82 are present at both end parts of the tubular member 32, and an interior space is isolated. Power needs to be supplied to the coil 22 of the

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stator 2 in this isolated interior space of the tubular member 32. Note that "isolated" here refers to a state of seemingly no location where the lead wire for supplying power can be routed from the outside to the inside.

In the rotating device 1 according to the present embodiment, passing the lead wire through the cavity within the shaft member 5 electrically connects the inside of the isolated internal space of the tubular member 32 to the outside of the internal space. Therefore, the lead wire can power the coil 22 of the stator 2 in the interior space of the tubular member 32.

Of the plurality of components included in the rotating device 1 configured as described above, a portion of the components (stator 2, rotor 3, bearing 4, inner impeller 81, spacer 82, and shaft member 5) form a rotational mechanism. This rotational mechanism rotates the rotor blades. With regard to this portion of components, the rotor 3 surrounding the stator 2 is rotatable with respect to the stator 2 fixed to the shaft member 5 and constitutes a so-called outer-rotor type brushless motor. In a typical outer-rotor type brushless motor, a shaft member fixed to a rotor rotates and the shaft member extracts a rotational force, whereas in the rotating device 1 according to the present embodiment, the shaft member 5 is a member at the fixed side, and is configured so that the rotational force is directly extracted from the rotor 3.

The housing 7 is a member having a cylindrical shape, and is formed of a plastic or a metal, for example. Although not illustrated, both ends in the axial direction of the housing 7 are openings (hereinafter, an opening at the upper side a is referred to as an "upper end opening" and an opening at the lower side b is referred to as a "lower end opening").

A portion of the components of the rotating device 1 are accommodated in an interior of the housing 7, and the shaft member 5 is fixed to an upper end part and a lower end part of the housing 7. The upper end part and the lower end part of the housing 7 each include three spoke parts 71a connected from a tubular main body part (tubular part 72) of the housing 7, and a disk part 71b coupled to the spoke parts 71a, and the shaft member 5 is fixed to the disk part 71b.

The housing 7 and the shaft member 5 constitute members at the fixed side. An upper end opening 75 and a lower end opening 76 are provided at the upper end part and the lower end part of the housing 7, and the upper end opening 75 and the lower end opening 76 each surround the disk part 71b and shaft member 5.

The rotor blades 6 are attached to the outer peripheral surface of the tubular member 32 of the rotor 3, across substantially the entire axial direction (axial line x direction) of the tubular member 32. The rotor blade 6 will now be described with reference to FIGS. 6 and 7.

FIG. 6 is a perspective view of only the rotor blade 6 extracted from the rotating device 1 according to the present invention, and FIG. 7 is a perspective view of the rotor blade 6, as viewed from near the upper side a in the axial direction.

The rotor blade 6 includes an inner tube 61 attached to the outer peripheral surface of the tubular member 32, an outer tube 63 spaced apart from and opposing an inner peripheral surface of the housing 7, and a plurality of (five in the present embodiment) blades 62 disposed between the inner tube 61 and the outer tube 63. A cavity 64 is formed between the inner tube 61 and the outer tube 63, and the blades 62 disposed in the cavity couple the inner tube 61 with the outer tube 63. Accordingly, the integrated rotor blade 6 is formed by the inner tube 61, the blades 62, and the outer tube 63.

The blades 62 are oriented in a direction inclined with respect to the axial line x direction of the shaft member 5.

The five blades **62** are equally spaced apart in the circumferential directions of and inclined at the same angle.

The rotor blade **6** is attached to the outer peripheral surface of the tubular member **32** by fitting the tubular member **32** of the rotor **3** into an inner peripheral part (inner hole) of the inner tube **61**, and fixing the tubular member **32** in a predetermined position by an adhesive or the like. Accordingly, the rotor blade **6** rotates in the housing **7** as the rotor **3** rotates.

The rotor blade **6** rotates in accordance with the rotation of the rotor **3** and thus rotates together with the rotor **3**, generating a flow of a gas in the axial line x direction in accordance with the rotation of the rotor blade **6**. This flow of air occurs in either an upward direction or a downward direction of the axial direction of the shaft member **5** in the cavity **64** between the inner tube **61** and the outer tube **63**. In the rotating device **1** according to the present embodiment, the rotating device **1** is configured to be driven to rotate the rotor blades **6** in the clockwise circumferential direction e, causing the air taken in from the lower end opening **76** to pass through the cavity **64** and be blown out from the upper end opening **75**.

The rotating device **1** according to the present embodiment is provided with the rotor blades **6** at an outer peripheral surface of the rotor **3** serving as a rotating body and is provided with the housing **7** having a tubular shape so as to surround the rotor blades **6**, and thus one of both end openings of the housing **7** is a suction port and the other is a discharge port, and the rotating mechanism (portion of the plurality of components of the rotating device **1**) rotating the rotor blades and the rotor blades **6** can be accommodated in the internal space of the housing **7**. One of the inner impeller **81** and the spacer **82** can be the suction port and the other can be the discharge port. In particular, because the rotor blades **6** are positioned in a flow channel (also referred to as a wind passage) where air flows, space can be saved and size reduction of the entire rotating device can be achieved.

In the rotating device **1** according to the present embodiment, the space communicating with the lower end opening **76** from the upper end opening **75** is a cavity so as not to inhibit the flow of air due to members other than the spoke parts **71a**. Furthermore, because the space has a straight tubular shape except for the space occupied by the cylindrical motor, air can flow straight. Thus, air can be fed out straight from the upper end opening toward the lower end opening by rotating the rotor blades **6**. Thus, according to the rotating device **1** according to the present embodiment, air can be efficiently fed out, and a supply of strong wind and large wind volume (large suction volume) can be achieved.

In a rotating device according to a configuration of a conventional motor with a rotating shaft member projecting from the motor, the rotating shaft member rotates with a side of the rotating shaft member being supported, and the rotational force is extracted from the projecting other end side. This easily causes deviation of rotation; however, in the rotating device **1** according to the present embodiment, the rotor **3** itself supported by the bearing **4** rotates as the rotating body, stabilizing the rotation of the rotor **3**.

In the rotating device **1** according to the present embodiment, because the first bearing **41** and the second bearing **42** are fixed respectively to both end parts of the rotor **3**, and the rotor **3** serving as the rotating body is supported, the rotation of the rotor **3** is stabilized with respect to the shaft member **5**. In particular, because the magnet **31** as a component of the rotor **3** serving as the rotating body and having a predetermined weight is disposed between the first bearing **41** and the second bearing **42** rotatably supporting the rotor **3** in the

axial direction of the shaft member **5**, the balance in the axial direction is improved and the rotation of the rotor **3** is stabilized.

Although the positions of the bearings are more preferably at both end parts of a rotating body as in the present embodiment, positions near both the end parts of the rotating body also makes the rotation of the rotating body with respect to the shaft member sufficiently stable. "Near" used herein refers to a position close to one of both end parts of the rotating body, between both end parts and a center part of the rotating body, and cannot be expressly defined by a numerical value. For example, a region within a length of 20% from one of both ends in the axial direction of the rotating body, or preferably, a region within a length of 10% from one of both the ends is included in the concept of "near both end parts".

In the rotating device **1** according to the present embodiment, the first bearing **41** and the second bearing **42** are members having the same configuration, improving a balance in the axial direction between rotating portions including the outer rings **41a**, **42a** as parts of the bearing **4** and the rotor **3**. That is, in the rotating device **1** according to the present embodiment, the balance in the axial direction of the entire rotating device **1** is improved, stabilizing the rotation of the rotor **3** from this perspective as well.

As described above, in the rotating device **1** according to the present embodiment, size reduction of the entire rotating device can be realized, deviation of rotation of the rotor **3** is unlikely to occur, and high precision stabilization can be achieved.

The stabilization of the rotation of the rotor **3** means that uneven rotation is less likely to occur, and thus the rotating device **1** can achieve high rotation and high torque. That is, according to the rotating device **1** according to the present embodiment, a device having excellent characteristics as a rotating device while achieving size reduction can be provided.

In the present embodiment, an example of a configuration with both upper and lower end parts of the shaft member **5** fixed to the housing **7** is given; however, it is sufficient that at least one end part or a portion of the shaft member **5** be fixed to the housing, as long as the shaft member **5** at the fixed side is fixed to the housing **7** in some manner.

As described above, the rotating device **1** of the present embodiment can realize high performance with a small size. However, because of the high performance, the coil **22** and the stator core **21** are more likely to become higher in temperature due to an increased speed of rotation, for example. This makes it necessary to cool the stator **2** including the heat generation source such as the coil **22** and the stator core **21**; however, ensuring sufficient cooling just by providing a ventilation port for cooling in the tubular member **32** may be difficult. In the first place, it is difficult to secure a space for the ventilation port. When a ventilation port is provided in the tubular member **32** to cool the stator **2**, the location for the ventilation port must be secured, sometimes hindering size reduction of the rotating device **1**.

In contrast, in the present embodiment, the coupling members (inner impeller **81** and spacer **82**) including the ventilation channel communicating with the inside and the outside of the rotor **3** are disposed between the bearing **4** and the rotor **3**, thereby achieving efficient cooling of the stator **2**.

FIG. **8** illustrates an explanatory diagram for describing a flow of a gas for cooling the inside of the rotor **3** of the rotating device **1** according to the present embodiment. Note that, in FIG. **8**, the housing **7** is omitted.

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As described above, by the action of the rotor blades **6**, a gas flow occurs. In the gas flow, the gas taken in from the lower end opening **76** passes through the cavity **64** and is blown out from the upper end opening **75**. Therefore, because a gas flow toward the upper side **a** occurs below the lower end opening **76** and above the upper end opening **75**, the gas introduced into this gas flow enters and exits into and out from the rotor **3**, even in the ventilation channel of the inner impeller **81** and spacer **82**, and a gas flow in the upper side **a** direction occurs.

Furthermore, the inner impeller **81** rotates by the rotation of the rotor **3** in an arrow **h** direction (clockwise direction **e**) in FIG. **8**, and the gas flow in the upper side **a** direction in the cavity **64** is strengthened by the blades **81c**. As a result, a strong gas flow occurs in an arrow **g** direction in FIG. **8**, and the stator **2** including the heat generation source, such as the coil **22** and the stator core **21**, in the tubular member **32** of the rotor **3** is efficiently cooled by this gas flow.

In particular, because the coupling member of the upper end opening **75** corresponding to the discharge port side of the gas is the inner impeller **81** including the blades **81c**, a large volume of the gas can be taken into the interior of the rotor **3** and discharged. Accordingly, according to the rotating device **1** of the present embodiment, the heat generation source, such as the coil **22** and the stator core **21**, can be very efficiently cooled.

As indicated by the arrow **g** in FIG. **8**, in the path of the gas flow from the spacer **82** to the inner impeller **81**, only the coil **22**, the stator core **21**, and the magnet **31** are present. Thus, without the presence of obstacles to the flow of the gas other than these cooling targets, the gas is efficiently supplied to these cooling targets. Furthermore, other than coming into contact with these cooling targets, the gas flows in the axial line direction **x**. Accordingly, according to the rotating device **1** according to the present embodiment, the gas used for cooling can be efficiently supplied to the cooling targets.

In the present embodiment, the coupling members (inner impeller **81** and spacer **82**) including a ventilation channel communicating with the inside and the outside of the rotor **3** are disposed between the bearing **4** and the rotor **3**, and thus a ventilation port for the intake and discharge of the gas need not be provided at the tubular member **32**. That is, because the coupling members including the ventilation channel can be disposed without increasing a length in the axial line **x** direction, a reduction in the size of the rotating device **1** can be realized.

In particular, when a ventilation port is provided at the peripheral surface (side surface) of the tubular member **32** (furthermore, the inner tube **61** of the rotor blade **6**), it is desirable to dispose the inner impeller near the ventilation port. Accordingly, in this case, the length in the axial line **x** direction needs to be long, and thus a length of the entire rotating device in the axial direction increases. However, in the present embodiment, an increase in the length of the entire rotating device in the axial direction can be suppressed.

When a ventilation port is provided at the peripheral surface of the tubular member **32** (furthermore, the inner tube **61** of the rotor blade **6**), the gas is taken in from a gas flow generated by the rotor blades **6** in the cavity **64** and the gas is returned to this gas flow, leading to disruption of the gas flow in the cavity **64**. However, in the present embodiment, the intake port (the spacer **82**) and the discharge port (inner impeller **81**) of the gas used for cooling are independent from the gas flow generated by the rotor blades **6** in the cavity **64**, and thus disruption of the gas flow caused by the

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rotor blades **6** can be suppressed, and the supply of a strong wind and a large wind volume (large suction volume) as a blowing device (or suction device) can be achieved.

In the present embodiment, of the two coupling members, the coupling member at the upper side **a** corresponding to the discharge port side of the gas is the inner impeller **81** including the blades **81c**. However, the coupling member at the lower side **b** corresponding to the intake port side of the gas may be the inner impeller including the blades **81c**, and the coupling member at the upper side **a** corresponding to the discharge port side may be replaced with a spacer and the coupling member at the lower side **b** corresponding to the intake port side may be replaced with the inner impeller. At least one of the discharge port side and the intake port side is an inner impeller, allowing a large volume of the gas to be taken into the rotor **3** and the cooling efficiency to be further improved. In particular, as in the present embodiment, the inner impeller **81** including the blades **81c** is provided as a coupling member, making it possible to cause the gas to flow through the coupling member to the stator **2** in the interior of the rotor **3** and cool the stator, and increase the wind volume emitted from the rotating device **1**.

The rotor blade **6** includes the inner tube **61** attached at the outer peripheral surface of the tubular member **32**, the outer tube **63**, and the plurality of blades **62** disposed between the inner tube **61** and the outer tube **63** and coupling the area between the inner tube **61** and the outer tube **63**. The rotor blade includes the outer tube **63** surrounding the plurality of blades **62**, making it possible to efficiently blow the gas from the rotating device **1**. Thus, the gas can be blown even if the housing **7** accommodating the rotor blades **6** is not included.

Second Embodiment

A rotating device according to a second embodiment as one example of the present invention will now be described. FIG. **9** is a transparent cross-sectional view of a cross section including the axial line **x** of a rotating device **201** according to the second embodiment. The rotating device **201** according to the second embodiment differs from the rotating device **1** according to the first embodiment in a configuration around the upper and lower bearings, and a structure of the rotor blade.

In the second embodiment, members or components having the same function and structure as the function and structure of the first embodiment are denoted in the drawings of the present embodiment by the same reference signs as the reference signs of the first embodiment, and detailed descriptions of those members and components will be omitted (unless a particular description is added).

A coupling member (hereinafter referred to as "spacer") **281** is disposed between the outer ring **41a** of the first bearing **41** and the inner peripheral surface of the end part of the tubular member **32** at the upper side **a**, and a coupling member (hereinafter referred to as "spacer") **282** is also disposed between the outer ring **42a** of the second bearing **42** and the inner peripheral surface of the end part of the tubular member **32** at the lower side **b**. These spacers **281**, **282** have the same shape as the shape of the spacer **82** in the first embodiment.

The rotating device **201** according to the present embodiment includes a blade member **291** adjacent to an outer side (upper side **a**) of the spacer **281** in the axial line **x** direction and including a plurality of blades (reference sign **291b** described below). Further, the rotating device **201** according to the present embodiment includes a blade member **292** adjacent to an outer side (lower side **b**) of the spacer **282** in

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the axial line x direction and including a plurality of blades (reference sign **292b** described below).

FIG. 10 illustrates an enlarged perspective view of only the blade member **291** at the upper side a in FIG. 9 extracted from the rotating device **201**. As illustrated in FIG. 10, the blade member **291** includes a ring part **291a**, a plurality of (in the present embodiment, six) blades **291b** extending in the center axis direction from the ring part **291a**, and a projecting part **291c** projecting downward from a coupling part side with the ring part **291a** of each blade **291b**.

FIG. 11 illustrates an enlarged perspective view of only the blade member **292** at the lower side b in FIG. 9 extracted from the rotating device **201**. As illustrated in FIG. 11, the blade member **292** includes a ring part **292a**, a plurality of (in the present embodiment, six) blades **292b** extending in the center axis direction from the ring part **292a**, and a projecting part **292c** projecting downward from a coupling part side with the ring part **292a** of each blade **292b**.

The blade member **291** and the blade member **292** are disposed vertically inverted but are members having substantially the same shape and the same configuration.

The blade members **291**, **292** may be molded with any material such as a resin or aluminum or other metal, but are preferably molded with a resin from the perspectives of weight reduction, low cost, and moldability.

As illustrated in FIGS. 10 and 11, the plurality of blades **291b**, **292b** is oriented in a direction inclined with respect to the axial line x direction (axial direction of the shaft member **5**). The six blades **291b**, **292b** are inclined at the same angle. Therefore, the blade members **291**, **292** are configured to function as a so-called fan for generating a flow of a gas in the axial line x direction by rotating.

FIG. 12 illustrates an enlarged perspective view of only the spacer **281** and the blade member **291** at the upper side a in FIG. 9 extracted from the rotating device **201**. The spacer **282** and the blade member **292** at the lower side b in FIG. 9 are also disposed vertically inverted but have the same shapes or structures, and thus the reference signs of the spacer **282** and the blade member **292** are also denoted in parentheses in FIG. 12.

FIG. 13 illustrates an exploded perspective view of the rotating device **201** according to the present embodiment. In FIG. 13, the housing **7** is not illustrated.

As illustrated in FIGS. 12 and 13, the blade members **291**, **292** are stacked on the spacers **281**, **282** so that the projecting parts **291c**, **292c** face the spacers **281**, **282**.

The spacers **281**, **282** includes ring parts (hereinafter referred to as "inner ring parts") **281a**, **282a** at an inner side, ring parts (hereinafter referred to as "outer ring parts") **281b**, **282b** at an outer side, and a plurality of (three in the present embodiment) spokes **281c**, **282c** connecting areas between the inner ring parts **281a**, **282a** and the outer ring parts **281b**, **282b**.

The spacers **281**, **282** have a simple hole opening shape to ensure accuracy and strength as coupling members with ease and at low cost.

An inner peripheral surface of the inner ring part **281a** of the spacer **281** is fixed to the outer peripheral surface of the outer ring **41a** of the first bearing **41**. An inner peripheral surface of the inner ring part **282a** of the spacer **282** is fixed to the outer peripheral surface of the outer ring **42a** of the second bearing **42**. The fixing of the inner peripheral surfaces of the inner ring parts **281a**, **282a** and the outer peripheral surfaces of the outer rings **41a**, **42a** is not particularly limited, and may be performed by any known method, such as, for example, press-fitting, interference-

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fitting, fixing with an adhesive, or engagement, or a combination of a plurality of these fixing means (fixing elements).

Outer peripheral surfaces of the outer ring parts **281b**, **282b** of the spacers **281**, **282** are fixed to the inner peripheral surfaces of the end parts of the tubular member **32** at both the upper and lower sides ab in the axial line x direction. The fixing of the outer peripheral surfaces of the outer ring parts **281b**, **282b** and the inner peripheral surface of the tubular member **32** is also not particularly limited, and may be performed by any known method, such as, for example, press-fitting, interference-fitting, fixing with an adhesive, or engagement, or a combination of a plurality of these fixing means (fixing elements).

The spacers **281**, **282** form penetrating parts **281d**, **282d** penetrating as a ventilation channel enabling the passage of a gas at positions excluding positions of the spokes **281c**, **282c** between the inner ring parts **281a**, **282a** and the outer ring parts **281b**, **282b**. The spokes **281c**, **282c** are connected to the inner ring parts **281a**, **282a** and the outer ring parts **281b**, **282b** at substantially centers of the spacers **281**, **282** in a thickness direction (axial line x direction). Therefore, in the spacers **281**, **282**, the spokes **281c**, **282c** are positioned inward of end surfaces of the inner ring parts **281a**, **282a** and the outer ring parts **281b**, **282b** at the stator **2** side. These spokes **281c**, **282c**, inner ring parts **281a**, **282a**, and outer ring parts **281b**, **282b** form stepped parts recessed in the axial line x direction.

The projecting parts **291c**, **292c** of the blade members **291**, **292** are fitted into and thus integrated with the penetrating parts and the stepped parts in the spacers **281**, **282**, as illustrated in FIG. 12. A plate spring **250** is attached near the end part of the shaft member **5** at the upper side a, biasing the blade member **291** and the bearing **41** from the upper side a. With this biasing force of the plate spring **250**, the blade member **291** and the spacer **281** are stably integrated.

A plate spring **251** is attached near the end part of the shaft member **5** at the lower side b, biasing the blade member **292** and the bearing **42** from the lower side b. With this biasing force of the plate spring **251**, the blade member **292** and the spacer **282** are stably integrated.

As described above, the spacers **281**, **282** are fixed to the inner peripheral surface of the tubular member **32** of the rotor **3**.

Accordingly, the blade members **291**, **292** fixed to the tubular member **32** of the rotor **3** with the spacers **281**, **282** interposed between the blade members **291**, **292** and the tubular member **32** are configured to rotate in accordance with the rotation of the rotor **3**.

Next, a rotor blade **206** will be described with reference to FIG. 14.

FIG. 14 is a perspective view removing only the housing **7** from the rotating device **201** according to the present embodiment. As illustrated in FIG. 14, the rotor blade **206** is attached to the outer peripheral surface of the tubular member **32** of the rotor **3**, extending across about 70% of the length of the tubular member **32** in the axial direction (axial line x direction).

The rotor blade **206** includes three fixing parts **261a**, **261b**, **261c** having an annular shape, arranged at equal intervals in the axial line x direction, and attached to the outer peripheral surface of the tubular member **32**, and a plurality of (six in the present embodiment) blades **262** fixed obliquely with respect to the axial line x direction so as to bridge these three fixing parts **261a**, **261b**, **261c**.

The blades **262** are oriented in a direction inclined with respect to the axial line x direction of the shaft member **5**.

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The six blades **262** are equally spaced apart in the circumferential directions of and inclined at the same angle.

The rotor blade **206** is attached to the outer peripheral surface of the tubular member **32** by fitting the tubular member **32** of the rotor **3** into the rings of the three fixing parts **261a**, **261b**, **261c**, and fixing the tubular member **32** in a predetermined position by an adhesive or the like. Accordingly, the rotor blade **206** rotates in the housing **7** as the rotor **3** rotates.

The rotor blade **206** rotates in accordance with the rotation of the rotor **3**, and thus rotates together with the rotor **3**, generating a flow of a gas in the axial line *x* direction in accordance with the rotation of the rotor blade **206**. This flow of air occurs in either the upward direction or the downward direction of the axial direction of the shaft member **5** in a cavity **277** between the rotor **3** and the housing **7**. In the rotating device **201** according to the present embodiment, the rotating device **201** is configured to be driven to rotate the rotor blade **206** in the clockwise circumferential direction *e*, causing the gas taken in from the lower end opening **76** to pass through the cavity **277** and blow out from the upper end opening **75**.

The rotating device **201** according to the present embodiment, as in the first embodiment, can achieve a size reduction of the entire rotating device, supply a strong wind and a large wind volume (large suction volume), stabilize the rotation of the rotor **3**, and achieve high rotation and high torque, and thus provide excellent characteristics as a rotating device.

In the present embodiment as well, the coupling members (spacers **281**, **282**) including the ventilation channel communicating with the inside and the outside of the rotor **3** are disposed between the bearing **4** and the rotor **3**, thereby achieving efficient cooling of the stator **2**.

FIG. **15** illustrates an explanatory diagram for describing a flow of a gas for cooling the inside of the rotor **3** of the rotating device **201** according to the present embodiment.

As described above, by the action of the rotor blade **206**, a gas flow occurs. In the gas flow, air taken in from the lower end opening **76** passes through the cavity **277** and is blown out from the upper end opening **75**. Therefore, because a gas flow toward the upper side *a* occurs below the lower end opening **76** and above the upper end opening **75**, the gas introduced into this gas flow enters and exits into and out from the rotor **3**, even in the ventilation channel of the spacers **281**, **282**, and a gas flow in the upper side *a* direction occurs.

Furthermore, by the rotation of the rotor **3** in the arrow *h* direction (clockwise direction *e*) in FIG. **15**, the blade members **291**, **292** rotate, strengthening the gas flow in the upper side *a* direction in the tubular member **32** by the blades **291b**, **292b**. As a result, a strong gas flow occurs in the arrow *g* direction in FIG. **15**, and the stator **2** including the heat generation source, such as the coil **22** and the stator core **21**, in the tubular member **32** of the rotor **3** is efficiently cooled by this gas flow.

In particular, the blade members **291**, **292** including the blades **291b**, **292b** are provided at both sides of the upper end opening **75** and the lower end opening **76**, and thus a large volume of the gas can be taken into the interior of the rotor **3** and discharged. Accordingly, according to the rotating device **201** of the present embodiment, the heat generation source, such as the coil **22** and the stator core **21**, can be very efficiently cooled.

As indicated by the arrow *g* in FIG. **15**, in the path of the gas flow from the spacer **282** to the spacer **281**, only the coil **22**, the stator core **21**, and the magnet **31** are present. Thus,

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without the presence of any obstacle to the flow of the gas other than these cooling targets, the gas is efficiently supplied to these cooling targets. Furthermore, other than coming into contact with these cooling targets, the gas flows in the axial line direction *x*. Accordingly, according to the rotating device **201** according to the present embodiment, the gas used for cooling can be efficiently supplied to the cooling targets.

In the present embodiment, the coupling members (spacers **281**, **282**) including the ventilation channel communicating with the inside and the outside of the rotor **3** are disposed between the bearing **4** and the rotor **3**, and thus a ventilation port for the intake and discharge of the gas need not be provided at the tubular member **32**. That is, because the coupling members including the ventilation channel can be disposed without increasing a length in the direction of the axial line *x* direction, a reduction in the size of the rotating device **201** can be realized.

When a ventilation port is provided at the peripheral surface of the tubular member **32**, the gas is taken in from a gas flow generated by the rotor blade **206** in the cavity **277** and the gas is returned to this gas flow, leading to disruption of the gas flow in the cavity **277**. However, in the present embodiment, the intake port (blade member **292** and spacer **282**) and the discharge port (spacer **281** and blade member **291**) of the gas used for cooling are independent from the gas flow by the rotor blade **6** in the cavity **277**, and thus disruption of the gas flow generated by the rotor blade **6** can be suppressed, and the supply of a strong wind and a large wind volume (large suction volume) as a blowing device (or suction device) can be achieved.

In the present embodiment, an example including the blade members **291**, **292** adjacent to both of the two coupling members (spacers **281**, **282**) and including the blades **291b**, **292b** is given; however, by providing a blade member at at least one of the discharge port side and the intake port side, it is possible to take a large volume of the gas into the rotor **3** and further improve the cooling efficiency. A blade member including blades may be provided at the discharge port side to improve the wind volume blown out from the rotating device and improve a wind pressure.

With the rotor blade **206** including the fixing parts **261a**, **261b**, **261c** having annular shapes, a region (area) coming into contact with the rotor (rotating body) **3** can be reduced and, as a result, a region (area) exposed to the gas at the outer peripheral side of the rotor **3** can be increased, making it possible to improve a heat dissipation of the rotor **3**.

In the present embodiment, unlike the first embodiment with the coupling member itself as the inner impeller **81**, the coupling members (spacers **281**, **282**) and the blade members **291**, **292** generating the gas flow in the tubular member **32** are separated.

Therefore, in the present embodiment, there is a high degree of freedom in the shape of the coupling member, making it possible to enlarge a cross-sectional area of the ventilation channel in the coupling member and improve the strength of the coupling member in comparison to the first embodiment.

The high degree of freedom in the shape of the coupling member significantly reduces size limitation of the bearing **4** side.

The rotating device can be assembled without the blade members and can be shaped in that state, and thus a plurality of types of blade members of different shapes, sizes, and thicknesses, for example, can be prepared for different applications and purposes, and selection can be made from among the different types on a case-by-case basis. That is, by

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mass producing, or the like, the rotating device in a state without the blade members and then assembling the blade members in accordance with the application or purpose as appropriate, it is possible to accommodate a wide range of needs and achieve a significant advantage in terms of cost.

Third Embodiment

A rotating device according to a third embodiment as one example of the present invention will now be described. FIG. 16 is a transparent cross-sectional view of a cross section including the axial line x of a rotating device 301 according to the third embodiment. The rotating device 301 according to the third embodiment differs from the rotating device 1 according to the first embodiment and the rotating device 201 according to the second embodiment in a configuration around the upper and lower bearings, and a structure of the rotor blade.

In the third embodiment, members or components having the same function and structure as the functions and structures of the first embodiment or the second embodiment are denoted in the drawings of the present embodiment by the same reference signs as the reference signs of the first embodiment or the second embodiment, and detailed descriptions of those members and components will be omitted (unless a particular description is added).

A coupling member (hereinafter referred to as “spacer”) 381 is disposed between the outer ring 41a of the first bearing 41 and the inner peripheral surface of the end part of the tubular member 32 at the upper side a, and a coupling member (hereinafter referred to as “spacer”) 382 is also disposed between the outer ring 42a of the second bearing 42 and the inner peripheral surface of the end part of the tubular member 32 at the lower side b.

The rotating device 301 according to the present embodiment includes a blade member 391 adjacent to an outer side (upper side a) of the spacer 381 in the axial line x direction and including a plurality of blades (reference sign 391c described below). The rotating device 301 according to the present embodiment includes a blade member 392 adjacent to an outer side (lower side b) of the spacer 382 in the axial line x direction and including a plurality of blades (reference sign 392c described below).

FIG. 17 illustrates an enlarged perspective view of only the spacer 381 and the blade member 391 at the upper side a in FIG. 16 extracted from the rotating device 301. FIG. 18 illustrates a cross-sectional perspective view taken along a cross section B-B including the axial line x in FIG. 17. Furthermore, FIG. 19 illustrates an exploded perspective view of the spacer 381 and the blade member 391.

The spacer 382 and the blade member 392 at the lower side b in FIG. 16 are also disposed vertically inverted but have substantially the same shapes and the same configurations. Accordingly, the reference signs of the spacer 382 and the blade member 392 are also denoted in parentheses in FIGS. 17 to 19. In the following, the spacer 381 and the blade member 391 at the upper side a will be mainly described, and the same applies to the spacer 382 and the blade member 392 at the lower side a.

As illustrated in FIGS. 17 to 19, the blade member 391 includes a ring part (hereinafter referred to as “inner ring part”) 391a at an inner side, a ring part (hereinafter referred to as “outer ring part”) 391b at an outer side, and the plurality of (three in the present embodiment) blades 391c connecting an area between the inner ring part 391a and the outer ring part 391b. With the ring parts (inner ring part 391a and outer ring part 391b) provided at both the inner periph-

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eral side and the outer peripheral side, the blades 391c are reinforced and the strength of the blade member 391 is improved.

The blade members 391, 392 may be molded with any material such as a resin or aluminum or other metal, but are preferably molded with a resin from the perspectives of weight reduction, low cost, and moldability.

As illustrated in FIGS. 17 to 19, the three blades 391c are oriented in a direction inclined with respect to the axial line x direction (axial direction of the shaft member 5). These three blades 391c are inclined at the same angle and are helical in shape. A cavity 391d is formed between the three blades 391c.

The blade member 391 rotates in one of the circumferential directions ef, moving a gas in one of the directions of a double-headed arrow k in FIGS. 17 and 18 in the cavity 391d, generating a flow of the gas in the axial line x direction. That is, the blade member 391 functions as a so-called fan.

As illustrated in FIG. 18, the inner ring part 391a includes a conical tubular part 391e including an inclined surface having a cross-sectional shape extending from the upper side a to the lower side b, and being thick at a lower side. The inner ring part 391a includes a projecting part (hereinafter referred to as “cylindrical projecting part”) 391f having a cylindrical shape and extending from an outermost periphery of the conical tubular part 391e to the lower side b at a bottom part 391g of the conical cylindrical part 391e.

As illustrated in FIG. 18, the outer ring part 391b includes a conical tubular part 391i including an inclined surface having a cross-sectional shape extending slightly from the upper side a to the lower side b, and being slightly thick at a lower side, and a projecting part (hereinafter referred to as a “cylindrical projecting part”) 391h having a cylindrical shape extending from an inner peripheral side to the lower side b and notched and recessed at a corner of an outer periphery of an end part at the lower side b (the notched location is referred to as “notch 391k” as a recessed part).

A length of the cylindrical projecting part 391f is longer than a length of the cylindrical projecting part 391h in the axial line x direction, and end parts (lower ends) of the cylindrical projecting part 391f and the cylindrical projecting part 391h at the lower side b are at the same positions.

The spacer 381 includes a base 381c having an annular shape and opposing the three blades 391c of the blade member 391, a ring part (hereinafter referred to as “inner ring part”) 381a having a tubular shape, connected at an inner peripheral side of the base 381c, and extending in upward and downward directions ab of the axial line x, and a ring part (hereinafter referred to as “outer ring part”) 381b having a tubular shape, connected at an outer peripheral side of the base 381c, and extending in the upward and downward directions ab of the axial line x. In the axial line x direction, the inner ring part 381a is longer than the outer ring part 381b.

The spacers 381, 382 have a simple hole opening shape to ensure accuracy and strength as the coupling member with ease and at low cost.

An inner peripheral surface of the inner ring part 381a of the spacer 381 is fixed to the outer peripheral surface of the outer ring 41a of the first bearing 41. An inner peripheral surface of the inner ring part 382a of the spacer 382 is fixed to the outer peripheral surface of the outer ring 42a of the second bearing 42. The fixing of the inner peripheral surfaces of the inner ring parts 381a, 382a and the outer peripheral surfaces of the outer rings 41a, 42a is not particularly limited, and may be performed by any known

method, such as, for example, press-fitting, interference-fitting, fixing with an adhesive, or engagement, or a combination of a plurality of these fixing means (fixing elements).

Outer peripheral surfaces of the outer ring parts **381b**, **382b** of the spacers **381**, **382** are fixed to the inner peripheral surfaces of the end parts of the tubular member **32** at both the upper and lower sides ab in the axial line x direction. The fixing of the outer peripheral surfaces of the outer ring parts **381b**, **382b** and the inner peripheral surface of the tubular member **32** is also not particularly limited, and may be performed by any known method, such as, for example, press-fitting, interference-fitting, fixing with an adhesive, or engagement, or a combination of a plurality of these fixing means (fixing elements).

In the spacer **381**, three hole parts **381d** having an elliptical shape forming an arc are formed in the base **381c**. The hole parts **381d** form a ventilation channel enabling the passage of a gas. The base **381c** is connected to the inner ring part **381a** and the outer ring part **382b** at the slightly lower side b of the spacers **381**, **382** in the thickness direction (axial line x direction). Therefore, in the spacers **381**, the base **381c** is positioned inward of the end surfaces of the inner ring part **381a** and the outer ring part **381b** at the stator **2** side. The base **381c**, the inner ring part **381a**, and the outer ring part **381b** form a stepped part recessed in the axial line x direction.

The cylindrical projecting part **391f** and the cylindrical projecting part **391h** of the blade member **391** are fitted into and thus integrated with the stepped part of the spacer **381**, as illustrated in FIG. **18**. That is, lower ends of the cylindrical projecting part **391f** and the cylindrical projecting part **391h** come into contact with a surface of the base **381c** at the upper side a, an upper end of the inner ring part **381a** comes into contact with the bottom **391g** of the conical tubular part **391e**, an upper end of the outer ring part **381b** comes into contact with the notch **391k** of the conical tubular part **391e**, and the blade member **391** and the spacer **381** are engaged and integrated.

The cavity **391d** in the blade member **391** integrated with the spacer **381** is connected to the hole parts **381d** corresponding to the ventilation channel of the spacer **381** to form the ventilation channel of the blade member **391**. That is, in FIGS. **17** and **18**, the flow of the gas in the cavity **391d** indicated by the double-headed arrow k connects with the flow of the gas at the hole parts **381d** indicated by a double-headed arrow m in FIG. **18**, forming a ventilation channel communicating with the inside and the outside of the rotor **3** as a whole.

As illustrated in FIG. **16**, the plate spring **250** is attached near the end part of the shaft member **5** at the upper side a, biasing the blade member **391** and the bearing **41** from the upper side a. With this biasing force of the plate spring **250**, the blade member **391** and the spacer **381** are stably integrated.

The plate spring **251** is attached near the end part of the shaft member **5** at the lower side b, biasing the blade member **392** and the bearing **42** from the lower side b. Due to the biasing force of the plate spring **251**, the blade member **392** and the spacer **382** are stably integrated.

As described above, the spacers **381**, **382** are fixed to the inner peripheral surface of the tubular member **32** of the rotor **3**.

Accordingly, the blade members **391**, **392** fixed to the tubular member **32** of the rotor **3** with the spacers **381**, **382** interposed between the blade members **391**, **392** and the

tubular member **32** are configured to rotate in accordance with the rotation of the rotor **3**.

Next, a rotor blade **306** will be described with reference to FIGS. **20** and **21**.

FIG. **20** is a perspective view of only the rotor blade **306** extracted from the rotating device **301** according to the present embodiment. FIG. **21** is an exploded perspective view of the rotor blade **306**. As illustrated in FIG. **16**, the rotor blade **306** is attached to a central region of the tubular member **32** of the rotor **3** in the axial direction (axial line x direction), at the outer peripheral surface of the tubular member **32**.

The rotor blade **306** includes a tube **363** having a cylindrical shape and a plurality of (five in the present embodiment) blades **362** obliquely fixed to an inner peripheral surface of the tube **363**.

The blades **362** are oriented in a direction inclined with respect to the axial line x direction of the shaft member **5**. The five blades **362** are equally spaced apart in the circumferential directions ef, inclined at the same angle, and helical in shape.

As illustrated in FIGS. **20** and **21**, the rotor blade **306** is formed with three rotor blade members **306a**, **306b**, **306c** lined in the axial line x direction. The rotor blade members **306a**, **306b**, **306c** respectively include tubular parts **363a**, **363b**, **363c**, and five blade parts **362a**, **362b**, **362c** projecting inward from an inner surface. These five blade parts **362a**, **362b**, **362c** are formed as ribs extending toward the axial line at the inner surface.

The three rotor blade members **306a**, **306b**, **306c** are stacked in the axial line x direction, aligning the tubular parts **363a**, **363b**, **363c** in series in the axial line x direction and forming the one tube **363**. Similarly, the three rotor blade members **306a**, **306b**, **306c** are stacked in the axial line x direction, aligning the respective five blade parts **362a**, **362b**, **362c** in series in the axial line x direction and forming the five blades **362**.

In the axial direction (axial line x direction) or the circumferential directions ef of the rotor **3**, respective end parts of the blade parts **362a**, **362b**, **362c** oppose the end parts of other blade parts. Specifically, in the axial line x direction or in the circumferential directions ef, the end part of the blade part **362b** at the upper side a opposes the end part of the blade part **362a** at the lower side b, and the end part of the blade part **362b** at the lower side b opposes the end part of the blade part **362c** at the upper side a.

By fixing one or all of these opposing end parts of the blade parts **362a**, **362b**, **362c** and the opposing upper and lower end parts of the tubular parts **363a**, **363b**, **363c** by a fixing means (fixing element) such as an adhesive, it is possible to achieve the integrated rotor blade **306**. The fixing means (fixing element) is not limited to fixing by an adhesive, and various fixing means (fixing elements) can be employed as appropriate.

In the axial direction (axial line x direction) or circumferential directions ef of the rotor **3**, for each of the five blades **362** of the rotor blade **306**, one blade **362** and the blade **362** adjacent to this one blade **362** oppose each other. Taking a blade **362-2** in FIG. **20** as a representative of description, a side surface of the blade **362-2** opposes a blade **362-3** adjacent in the clockwise circumferential direction e or the axial line x direction at a lower end **362-2g** side. The side surface of the blade **362-2** opposes a blade **362-1** adjacent in the counterclockwise circumferential direction f or the axial line x direction at an upper end **362-2p** side.

In the axial line x direction or the circumferential directions ef, each end part of the five blades **362** of the rotor

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blade **306** is positioned between both end parts of the blade **362** adjacent to that blade **362**. Taking a blade **362-1** in FIG. **20** as a representative of description, a lower end **362-1g** of blade **362-1** is positioned between both of the end parts **362-2p**, **362-2g** of the blade **362-2** adjacent in the axial line *x* direction or the clockwise circumferential direction *e* of the blade **362-1**. Taking a blade **362-3** in FIG. **20** as a representative of description, an upper end **362-3p** of the blade **362-3** is positioned between both of the end parts **362-2p**, **362-2g** of the blade **362-2** adjacent in the axial line *x* direction or the counterclockwise circumferential direction *f* of the blade **362-3**.

In the axial line *x* direction, each end part of the five blades **362** of the rotor blade **306** opposes a portion (side surface) between both end parts of the blade **362** adjacent to that blade **362**. In other words, portions of the five blades **362** of the rotor blade **306** overlap each other in the axial line *x* direction. Therefore, the gas passes through the path formed between the side surfaces of two blades **362** adjacent to each other in the axial line *x* direction or the circumferential directions *ef*.

In the case of a shape of the circumferential direction of positions of the blades **362** overlapping each other, as with the rotor blade **306** in the present embodiment, molding is relatively difficult. However, in the present embodiment, the rotor blade **306** is divided into three rotor blade members **306a**, **306b**, **306c**. Accordingly, the individual rotor blade members **306a**, **306b**, **306c** have favorable moldability and can be easily molded even with a typical mold. Therefore, by stacking these rotor blade members **306a**, **306b**, **306c**, it is possible to form the rotor blade **306** having a relatively complex shape with ease and at low cost.

An opening angle of each of the blade parts **362a**, **362b**, **362c** of the rotor blade members **306a**, **306b**, **306c** (circumferential angle of the arc between blades parts adjacent to each other on the circumference along the blade part about the axial line *x* when viewed from one side (upper side *a*, for example) in the axial line *x* direction) is $180^\circ/m$ (where *m* is a positive integer; the number of blade parts) and, because there are five blade parts ($m=5$), one opening angle is 36° .

At outer peripheral surfaces of each of the tubular parts **363a**, **363b**, **363c** of the rotor blade members **306a**, **306b**, **306c**, alignment grooves **364** are provided at intervals of a center angle of 36° with half of the alignment grooves **364** coinciding with the locations of the blade parts **362a**, **362b**, **362c** at the inner peripheral side of the upper side *a*. These alignment grooves **364** are arranged alternately in locations with and locations without the blade parts **362a**, **362b**, **362c** at the inner peripheral side of the upper side *a*. Then, the three rotor blade members **306a**, **306b**, **306c** are stacked, connecting the blade parts **362a**, **362b**, **362c** to form the blade **362** having a helical shape.

The rotor blade **306** is attached to the outer peripheral surface of the tubular member **32** by fixing the end parts of the five blades **362** at the axial line *x* side in the radial direction at predetermined positions of the tubular member **32** of the rotor **3** with an adhesive or the like. Accordingly, the rotor blade **306** rotates in the housing **7** with rotation of the rotor **3**.

The rotor blade **306** rotates in accordance with the rotation of the rotor **3**, and thus rotates together with the rotor **3**, generating a flow of a gas in the axial line *x* direction in accordance with the rotation of the rotor blade **306**. This flow of air occurs in either the upward direction or the downward direction of the axial direction of the shaft member **5** in a cavity **377** between the rotor **3** and the housing **7**.

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As illustrated in FIG. **16**, in the rotating device **301** according to the present embodiment, when the rotating device **301** is driven to rotate the rotor blade **306** in the clockwise circumferential direction *e*, a gas flow occurs in an arrow *i* direction, and the gas taken in from the lower end opening **76** passes through the cavity **277** and is blown out from the upper end opening **75**. On the other hand, when the rotor blade **306** is rotated in the counterclockwise circumferential direction *f*, a gas flow occurs in an arrow *j* direction, and the gas taken in from the upper end opening **75** passes through the cavity **277** and is blown out from the lower end opening **76**.

The rotating device **301** according to the present embodiment, as in the first embodiment and the second embodiment, can achieve a size reduction of the entire rotating device, supply a strong wind and a large wind volume (large suction volume), stabilize the rotation of the rotor **3**, and achieve high rotation and high torque, and thus provide excellent characteristics as a rotating device.

In the present embodiment as well, the coupling members (spacers **381**, **382**) including the ventilation channel communicating with the inside and the outside of the rotor **3** are disposed between the bearing **4** and the rotor **3**, thereby achieving efficient cooling of the stator **2**.

As described above, by the rotation of the rotor **3** in the clockwise *e* direction or the counterclockwise *f* direction, the rotor blade **306** rotates, generating a gas flow (in the arrow *i* direction or the arrow *j* direction) of the air taken in from the lower end opening **76** or the upper end opening **75** passing through the cavity **377** and being blown out from the upper end opening **75** or the lower end opening **76** (refer to FIG. **16**).

The rotation of the rotor **3** in the clockwise *e* direction or the counterclockwise *f* direction causes the blade members **391**, **392** to rotate, and the blades **391c**, **392b** strengthen the gas flow in the upper side *a* direction or the lower side *b* direction in the tubular member **32**. As a result, the strong gas flow in the tubular member **32** of the rotor **3** efficiently cools the stator **2** including the heat generation source, such as the coil **22** and the stator core **21**.

In particular, the blade members **391**, **392** including the blades **391c**, **392b** are provided at both sides of the upper end opening **75** and the lower end opening **76**, and thus a large volume of the gas can be taken into the interior of the rotor **3** and discharged. Thus, the rotating device **301** of the present embodiment can extremely efficiently cool the heat generation source, such as the coil **22** and the stator core **21**.

These blade members **391**, **392** provided at the top and bottom include the blades **391c**, **392b** having vertically symmetrical helical shapes, making it possible to reverse the driving of the rotating device **301** to reverse the rotation direction of the rotor **3**, and thus generate a gas flow in both the upward and downward directions (arrow *i* direction and arrow *j* direction).

The direction of the gas flow (downward direction or upward direction) caused by the rotation of the blade members **391**, **392** switched in accordance with the rotation direction of the rotor **3** (clockwise direction *e* or counterclockwise direction *f*) is the same as the direction of the gas flow (arrow *i* direction and arrow *j* direction) generated by the rotation of the rotor blade **306** similarly switched. Accordingly, the total wind volume can be improved.

As in the second embodiment, the path of the gas flow in the tubular member **32** is free of obstructions to the gas flow other than the cooling targets, and a ventilation port for taking in and discharging the gas does not need to be provided in the tubular member **32**. Thus, as in the second

embodiment, the heat generation source can be efficiently cooled and the size of the rotating device **301** can be reduced.

In the present embodiment, the rotor blade **306** is fixed in contact with the outer peripheral surface of the tubular member **32** of the rotor **3** at the end parts of the five blades **362** at the axial line x side, making it possible to reduce a region (area) coming into contact with the rotor (rotating body) **3**. As a result, a region (area) exposed to the gas at an outer peripheral side surface of the rotor **3** can be increased, making it possible to improve the heat dissipation of the rotor **3**.

In the present embodiment, as in the second embodiment, the coupling members (spacers **381**, **382**) and the blade members **391**, **392** generating the gas flow in the tubular member **32** are separated. Therefore, the present embodiment is similar to the second embodiment in a high degree of freedom in the shape of the coupling member, allowing for an enlarged cross-sectional area of the ventilation channel in the coupling member, an improved strength of the coupling member, and a significantly reduced size limitation of the bearing **4** side.

The actions and effects resulting from being able to assemble the rotating device without the blade members are similar to the second embodiment.

In the above, the rotating device according to the present invention has been described with reference to preferred embodiments; however, the rotating device according to the present invention is not limited to the configurations of the embodiments described above. For example, in the embodiments described above, an example of one coupling member being the inner impeller **81** (first embodiment) and an example of the blade members **291**, **292** or the blade members **391**, **392** being provided at both sides of the upper end opening **75** and the lower end opening **76** (the second embodiment and the third embodiment) have been described; however, the configuration may not include any blades for generating a gas flow for cooling the inside of the rotor.

That is, even with the inner impeller **81** replaced with the spacer **82** in the first embodiment and the blade members **291**, **292** or the blade members **391**, **392** removed in the second embodiment and the third embodiment, the coupling member including the ventilation channel communicating with the inside and the outside of the rotating body (rotor **3**) is included, and thus the actions or the effects of the present invention can be expected.

Even without the blades for generating the gas flow for cooling the inside the rotor, a gas flow occurs. In this gas flow, the air taken in from the lower end opening **76** or the upper end opening **75** passes through the tubular member **32** and is blown out from the upper end opening **75** or the lower end opening **76**. Therefore, because a gas flow toward the upper side a or toward the lower side b occurs below the lower end opening **76** and above the upper end opening **75**, the gas introduced into this gas flow enters and exits into and out from the rotor **3**, even in the ventilation channel of the coupling member not including a blade, and a gas flow in the upper side a direction or the lower side b direction occurs.

In the embodiments described above, an example of a configuration with both upper and lower end parts of the shaft member **5** fixed to the housing **7** is given; however, it is sufficient that at least one end part or a portion of the shaft member **5** be fixed to the housing, as long as the shaft member **5** at the fixed side is fixed to the housing **7** in some manner.

In the embodiments described above, an example of a configuration with the shaft member **5** fixed to the disk part **71b** of the housing **7** is given; however, in the rotating device of the present invention, a housing **407** and shaft members **405**, **405'**, **405''** may be fixed via connection members **410**, **410'**, **410''**, as in modification examples illustrated in FIGS. **22** to **24**. The connection members **410**, **410'**, **410''** may be molded with any material such as a resin or aluminum or other metal.

In the modification examples illustrated in FIGS. **22** to **24**, the connection members **410**, **410'**, **410''** include tubular parts **410a**, **410a'**, **410a''** and flange parts **410b**, **410b'**, **410b''** having an annular shape. Inner peripheral surfaces of the tubular parts **410a**, **410a'**, **410a''** are fixed to outer peripheral surfaces of the shaft members **405**, **405'**, **405''** at end parts at the upper side a of the shaft members **405**, **405'**, **405''**. The flange parts **410b**, **410b'**, **410b''** project from outer peripheral surfaces of the tubular parts **410a**, **410a'**, **410a''** to the outer peripheral side c. Surfaces of the flange parts **410b**, **410b'**, **410b''** at the upper side a are at the lower side b of surfaces of the tubular parts **410a**, **410a'**, **410a''** at the upper side a. In the axial direction of the shaft members **405**, **405'**, **405''**, portions of the tubular parts **410a**, **410a'**, **410a''** are between the surfaces of the flange parts **410b**, **410b'**, **410b''** at the upper side a and the surfaces of the tubular parts **410a**, **410a'**, **410a''** at the upper side a. An inner peripheral surface of an annular part **471b** of the housing **407** comes into contact with outer peripheral surfaces of portions of the tubular parts **410a**, **410a'**, **410a''**, and the surfaces of the flange parts **410b**, **410b'**, **410b''** at the upper side a and a surface of the annular part **471b** of the housing **407** at the lower side b come into contact, engaging and fixing the connection members **410**, **410'**, **410''** and the housing **407**.

In the modification example illustrated in FIG. **22**, in the axial direction, the tubular part **410a** of the connection member **410** extends to an end part of a first bearing **441** at the lower side b. The tubular part **410a** is between the shaft member **405** and the first bearing **441** in the radial direction. An inner ring **441b** of the first bearing **441** is fixed to another portion of the tubular part **410a** at the lower side b with respect to the flange part **410b**.

In the tubular part **410a**, a plurality of through-holes **410c** having a circular shape and penetrating in the axial direction is lined in the circumferential direction at an equal interval. Each of the plurality of through-holes **410c** extends from an end part of the tubular part **410a** at the upper side a to an end part of the tubular part **410a** at the lower side b in the axial direction. In the modification example illustrated in FIG. **22**, a plurality of (for example, three) lead wires **483** connected to the coil **22** is drawn to the outside of the rotating device through one or more of the through-holes **410c**.

In the modification example illustrated in FIG. **23**, a surface of the tubular part **410a'** of the connection member **410'** at the lower side b and a surface of the flange part **410b'** at the lower side b form a plane. In the axial direction of the shaft member **405'**, the surface of the tubular part **410a'** at the lower side b and the surface of the flange part **410b'** at the lower side b oppose the surface of the first bearing **41** at the upper side a, and are separated by a predetermined interval from the surface of the first bearing **41** at the upper side a.

The shaft member **405'** is partially or entirely hollow in the axial direction, and is open at the upper side a. In the shaft member **405'**, a plurality (for example, three) openings **405a'** are provided at the lower side b of the first bearing **41** in the axial direction. Only one opening **405a'** may be provided. In the modification example illustrated in FIG. **23**,

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a plurality (for example, three) of lead wires **483'** connected to the coil **22** are configured to be drawn from the openings **405a'** into the interior of the shaft member **405'** and drawn from the end opening of the shaft member **405'** to the outside of the rotating device.

In the modification example illustrated in FIG. **24**, a surface of the tubular part **410a''** of the connection member **410''** at the lower side **b** and a surface of the flange part **410b''** at the lower side **b** form a plane. In the axial direction of the shaft members **405**, **405'**, **405''**, the surface of the tubular part **410a'** at the lower side **b** and the surface of the flange part **410b'** at the lower side **b** oppose the surface of the first bearing **41** at the upper side **a**, and are separated by a predetermined interval from the surface of the first bearing **41** at the upper side **a**.

In the outer peripheral surface of the shaft member **405''**, a plurality of (for example, three) recessed parts **405a''** extending linearly from an end part at the upper side **a** to the lower side **b** in the axial direction are formed radially at an equal interval. Each recessed part **405a''** extends to the lower side **b** of the first bearing **41**. Only one recessed part **405a''** may be provided. In the modification example illustrated in FIG. **24**, a plurality of (for example, three) lead wires **483''** connected to the coil **22** are configured to be drawn to the outside of the rotating device through one or more of the recessed parts **405a''**.

According to the modification examples illustrated in FIGS. **22** to **24**, the lead wires can be drawn out without difficulty from inside the rotating device while keeping the number of components at a minimum.

In the modification examples illustrated in FIGS. **22** to **24**, an example of a configuration with the connection members **410**, **410'**, **410''** being provided at the end part of the rotating device at the upper side **a** is given; however, the connection members **410**, **410'**, **410''** may be provided at the end parts of the rotating device at the lower side **b** or may be provided at both end parts. In the modification example illustrated in FIG. **23**, an example of a configuration with the shaft member **405a** being open at the upper side **a** is given; however, the shaft member **405'** may be open at the lower side **b**. Further, in the modification example illustrated in FIG. **24**, an example of a configuration with the recessed part **405a''** extending from the end part of the shaft member **405''** at the upper side **a** to the lower side **b** in the axial direction is given; however, the recessed part **405a''** may be formed extending from the end part of the shaft member **405''** at the lower side **b** to the upper side **a**. That is, in the modification examples illustrated in FIGS. **22** to **24**, an example of a configuration with the lead wires being drawn from the upper side **a** of the rotating device is given; however, a configuration with the lead wires being drawn out from the lower side **b** of the rotating device may be adopted.

In the first embodiment, an example of a configuration with both the inner ring part **82a** and the outer ring part **82b** of the spacer **82** having a cylindrical shape is given; however, as in the modification example illustrated in FIG. **25**, in the rotating device of the present invention, an inner ring part **582a** of a spacer **582** may include, at an end part at the lower side **b**, a projecting part **582aa** having an annular shape and projecting to the inner peripheral side **d**. An outer ring part **582b** of the spacer **582** may include, at the end part of the lower side **b**, a projecting part **582ba** having an annular shape and projecting to the outer peripheral side **c**. Here, the projecting part **582aa** engages with an end part of the inner ring **42b** of the second bearing **42** at the lower side **b** in the axial direction, and the projecting part **582ba**

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engages with the end part of the tubular member **32** at the lower side **b** in the axial direction.

According to the modification example illustrated in FIG. **25**, the projecting part **582aa** and the projecting part **582ba** are stoppers, facilitating the positioning of the spacer **582**, the rotor **3**, and the second bearing **42**. Further, misalignment caused by the spacer **582**, the rotor **3**, and the second bearing **42** mutually sliding in the axial direction can be prevented.

In the modification example illustrated in FIG. **25**, an example of a configuration with the projecting parts provided at the inner peripheral side **d** and the outer peripheral side **c** at the end part at the lower side **b** of the coupling member at the lower side **b** is given; however, the projecting parts may be provided at the inner peripheral side **d** and the outer peripheral side **c** at the end part of the upper side **a** of the coupling member at the upper side **a**, or both may be provided. The coupling member including the projecting part may be a spacer, or may be an inner impeller.

In the embodiments described above, the blowers or intake devices provided with the rotor blades **6**, **206**, **306** are given as examples; however, the rotating device of the present invention may use the rotational force for purposes other than rotation of the rotor blade. In this case, the rotor blade is not required, and a transmission mechanism such as a gear or a shaft for transmitting the rotational force to an intended target may be provided, or the rotating body may be configured to act directly on the intended target.

The rotating device of the embodiments described above may use a liquid, such as water or oil, instead of a gas. Therefore, in the rotating device in the embodiments described above, fluid including a gas or a liquid can be applied.

In addition, the rotating device according to the present invention may be appropriately modified by a person skilled in the art according to conventionally known knowledge. Such modifications are of course included in the scope of the present invention as long as these modifications still include the configuration of the present invention.

REFERENCE SIGNS LIST

1 Rotating device; **2** Stator; **3** Rotor (rotating body); **4** Bearing; **5** Shaft member; **6** Rotor blade; **7** Housing; **21** Stator core; **22** Coil; **23** Magnetic pole part; **24** Annular part; **21** Magnet; **32** Tubular member; **41** First bearing (bearing); **41a**, **42a** Outer ring; **41b**, **42b** Inner ring; **41c**, **42c** Bearing ball; **42** Second bearing (bearing); **61** Inner tube; **62** Blade; **63** Outer tube; **64** Cavity; **71a** Spoke part; **71b** Disk part; **72** Tubular part; **77** Space; **81** Inner impeller (coupling member); **81a**, **82a** Inner ring part; **81b**, **82b** Outer ring part; **81c** Blade; **82** Spacer (coupling member); **82c** Spoke; **201** Rotating device; **206** Rotor blade; **261a**, **261b**, **261c** Fixing part; **262** Blade; **277** Cavity; **281**, **282** Spacer (coupling member); **281a**, **282a** Inner ring part; **281b**, **282b** Outer ring part; **281c**, **282c** Spoke; **291** Blade member; **291a** Ring part; **291b** Blade; **291c** Projecting part; **292** Blade member; **292a** Ring part; **292b** Blade; **292c** Projecting part; **301** Rotating device; **306** Rotor blade; **306a**, **306b**, **306c** Rotor blade member; **362**, **362-1**, **362-2**, **362-3** Blade; **362a**, **362b**, **362c** Blade part; **362-1g**, **362-2g** Lower end; **362-2p**, **362-3p** Upper end; **363** Tube; **363a**, **363b**, **363c** Tubular part; **364** Alignment groove; **377** Cavity; **381**, **382** Spacer (coupling member); **381a**, **382a** Inner ring part; **381b**, **382b** Outer ring part; **381c**, **382c** Base; **381d**, **382d** Base; **391** Blade member; **391a** Inner ring part; **391b** Outer ring part; **391c** Blade; **391d** Cavity; **391e** Conical tubular part; **391f** Cylindrical projecting part; **391g** Bottom; **391h** Cylindrical projecting part; **391i** Conical

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tubular part; **391k** Notch; **392** Blade member; **392a** Ring part; **392b** Outer ring part; **392c** Blade; **392d** Cavity; **392e** Conical tubular part; **392f** Cylindrical projecting part; **392g** Bottom; **392h** Cylindrical projecting part; **392i** Conical tubular part; **392k** Notch; **405**, **405'**, **405"** Shaft member; **405a** Opening; **405a"** Recessed part; **407** Housing; **410**, **410'**, **410"** Connection member; **410a**, **410a'**, **410a"** Tubular part; **410b**, **410b'**, **410b"** Flange part; **410c** Through-hole; **441** First bearing (bearing); **441a** Outer ring; **441b** Inner ring; **441c** Bearing ball; **471b** Annular part; **483**, **483'**, **483"** Lead wire; **582** Spacer (coupling member); **582a** Inner ring part; **582b** Outer ring part; **582c** Spoke; **582aa**, **582ba** Projecting part

The invention claimed is:

1. A rotating device, comprising:
 - a shaft member;
 - a rotating body having a tubular shape and rotatable about the shaft member;
 - a bearing supporting the rotating body with respect to the shaft member; and
 - a coupling member disposed between the bearing and the rotating body in a radial direction, and
 - a blade member adjacent to the coupling member in an axial direction and including a plurality of blades, wherein
 - the blade of the blade member is inclined,
 - the coupling member includes a ventilation channel communicating with an inside and an outside of the rotating body,
 - the blade member includes a projecting part projecting toward a side of the coupling member, and
 - the projecting part fits into the coupling member.
2. The rotating device according to claim 1, comprising:
 - a housing having a tubular shape surrounding the rotating body; and
 - a stator inside the rotating body.
3. The rotating device according to claim 1, wherein the coupling member includes a plurality of blades, and the plurality of blades is oriented in a direction inclined with respect to an axial direction of the shaft member.
4. The rotating device according to claim 3, comprising bearings including the bearing, and coupling members including the coupling member, wherein
 - the bearings are provided at end parts of the rotating body or near the end parts of the rotating body in the axial direction of the shaft member,
 - the coupling members are disposed between the bearings and the rotating body,
 - one of the coupling member includes the blades.
5. The rotating device according to claim 4, wherein the other of the coupling members includes a plurality of spokes extending inward and outward in the radial direction.
6. The rotating device according to claim 1, wherein
 - a gas can flow inside the rotating body in the axial direction, and
 - the coupling member at a discharge port side of the gas includes the blades.
7. The rotating device according to claim 1, wherein the blade of the blade member is oriented in a direction inclined with respect to an axial direction of the shaft member.

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8. The rotating device according to claim 1, wherein the coupling member includes a plurality of spokes extending inward and outward in the radial direction.

9. The rotating device according to claim 8, wherein

- a gas can flow inside the rotating body in the axial direction, and

the blade member is provided at an outer side of the coupling member at a discharge port side of the gas in the axial direction of the rotating body.

10. The rotating device according to claim 1, wherein a rotor blade is provided at an outer peripheral surface of the rotating body.

11. The rotating device according to claim 10, wherein

- the rotor blade includes a plurality of blades each including a plurality of blade parts lined in the axial direction of the rotating body, and

each of the plurality of blades of the rotor blade opposes a blade adjacent to each of the plurality of blades of the rotor blade in a circumferential direction of the rotating body.

12. The rotating device according to claim 11, wherein an end part of each of the plurality of blades of the rotor blade is positioned between both end parts of the blade adjacent to each of the plurality of blades of the rotor blade in the circumferential direction of the rotating body.

13. The rotating device according to claim 1, wherein

- the bearings are provided at a end part of the rotating body or the end part of the rotating body in an axial direction of the rotating body,

the coupling members are disposed between the bearings and the rotating body, and the blade member is provided at an outer side of one of the coupling members in the axial direction of the rotating body.

14. A rotating device, comprising:

- a shaft member;
- a rotating body having a tubular shape and rotatable about the shaft member;

- a bearing supporting the rotating body with respect to the shaft member;

- a coupling member disposed between the bearing and the rotating body in a radial direction; and

- a blade member adjacent to the coupling member in an axial direction and including a plurality of blades, wherein

- the coupling member includes a first ventilation channel communicating with an inside and an outside of the rotating body,

- the blade member includes a second ventilation channel connected to the ventilation channel of the coupling member,

- a gas can flow inside the rotating body in both directions of axial direction, and

- the coupling member includes a base opposing the blade of the blade member in an axial direction of the shaft member.

15. The rotating device according to claim 14, wherein a hole part serving as the first ventilation channel is formed in the base.

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